
Shunting Concentrates Path Gains and Improves Local Credit Assignment in Dendritic Networks

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Abstract

1 Exact gradients for conductance-based dendritic trees factorize into a synapse-
2 local eligibility term and a single non-local compartment error; this factorization
3 is the foundation for local credit assignment in these networks. How well a so-
4 matic broadcast can substitute for the true compartment error depends on how
5 concentrated the conductance-stage path gains are across the tree. Across inhibited
6 noise-resilience regimes, an oracle experiment—replacing the broadcast with the
7 exact conductance-stage transported compartment errors—raises additive local
8 learning from roughly 38–43% to 99–100%, nearly matching shunting (98–99%),
9 strongly supporting path-gain fidelity as the dominant bottleneck rather than other
10 architectural differences. Under the default shared rank-1 broadcast used by the
11 headline feedforward architectures, shunting improves both gradient fidelity and
12 end-task learning relative to additive integration in the inhibited regimes where
13 conductance normalization matters: on MNIST, cosine alignment rises from 0.006
14 to 0.202, and on noise resilience the shunting advantage is largest once inhibition
15 is present. We further analyze structured rank- K feedback as the natural extension
16 when rank-1 broadcasts lose pathway identity. Together, these results identify
17 conductance-stage path-gain concentration via shunting inhibition as the key bio-
18 physical mechanism that determines when simple shared local credit assignment
19 works in conductance-based dendritic networks.

20 1 Introduction

21 Credit assignment in deep networks usually depends on backpropagation: global error transport
22 through symmetric pathways with no obvious biological substrate. Dendritic compartments suggest an
23 alternative. Each synapse has access to rich local state such as driving force, membrane conductance,
24 and branch voltage, while supervisory information could arrive as a low-bandwidth broadcast from
25 the soma [14]. The question here is when such local information is sufficient for useful learning.

26 Starting from conductance-based dendritic voltage equations [1], we derive exact gradients for
27 dendritic trees (Theorem 1) and show that each synaptic gradient factorizes into purely local eligibility
28 terms and a single non-local compartment error. Approximating that compartment error with a
29 broadcast signal yields a theorem-facing local rule (3F), together with practical heuristic wrappers
30 (4F/5F), and higher-rank broadcast extensions for tasks where different branches must be credited
31 differently.

32 The central finding is mechanistic: shunting improves broadcast fidelity by concentrating conductance-
33 stage path gains. On MNIST, conductance-stage path-gain CV stays at 0.21–0.27 under shunting
34 versus 0.53–1.08 under additive integration, so one shared broadcast is a much better proxy for
35 compartment-specific errors under shunting. On the noise-resilience sweep, per-soma broadcast
36 reaches cosine alignment of about 0.37–0.47 under shunting across the inhibited regimes, while

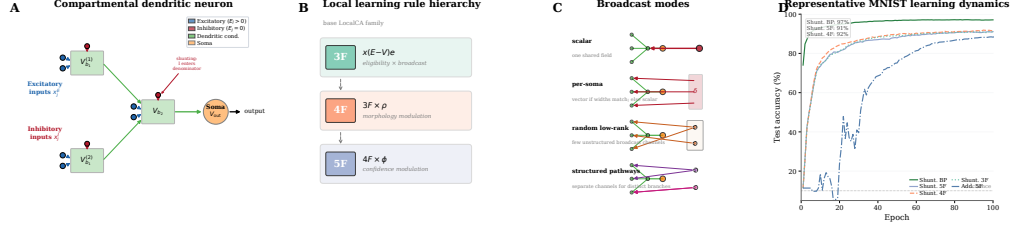


Figure 1: Model and credit assignment. (A) A compartmental dendritic neuron with excitatory (blue triangles) and inhibitory (red circles; shunting) synapses. Voltages propagate to the soma through learned dendritic conductances (green arrows). A somatic broadcast error e_n (dashed red) reaches all compartments; each synapse combines e_n with local quantities (presynaptic drive x_j , input resistance R_n^{tot} , and driving force $E_j - V_n$). (B) Compact LocalCA hierarchy: 3F is the theorem-facing local rule; 4F adds the heuristic morphology-aware factor ρ ; 5F further adds the heuristic confidence-like factor ϕ . (C) Broadcast-mode schematic. Scalar broadcast sends one shared field to all branches; per-soma broadcast sends a vector-valued somatic error only when output and layer widths match and otherwise falls back to scalar; random low-rank broadcast adds a small number of unstructured channels; structured pathway feedback uses separate channels to preserve pathway identity when different branches should receive different teaching signals. (D) Representative MNIST learning dynamics from the fixed-epoch Figure 1 sweep. Shunting with the stronger local wrappers learns rapidly and remains much closer to backprop than the additive control, making the rule hierarchy concrete before the paper turns to the more precise gradient-fidelity diagnostics in Fig. 2.

additive becomes negative once inhibition is introduced even though it can still perform well at $N_I=0$. Our question is therefore not whether local learning can match backpropagation on standard benchmarks, but what biophysical mechanism determines when low-bandwidth local credit assignment works in conductance-based dendritic networks. An oracle experiment that supplies the exact conductance-stage transported compartment errors strongly supports path-gain fidelity as the dominant bottleneck: across the inhibited noise-resilience regimes, additive local learning rises to essentially ceiling under the oracle, nearly matching shunting.

We quantify these effects with a gradient-fidelity diagnostic based on directional alignment and scale mismatch between local and backprop gradients (Fig. 2). This is a mechanism paper, not a benchmark-leadership paper: the paper’s contributions are exact factorization for conductance-based dendritic trees, a mechanistic account of shunting as conductance-stage path-gain concentration, direct fidelity and oracle experiments that isolate the broadcast bottleneck, and a structured rank- K viewpoint for settings where rank-1 broadcast fails.

2 Compartmental Voltage Model and Gradient Derivation

We use a steady-state conductance model derived from discretized passive cable dynamics [1, 2]. In normalized units, leak and inhibitory reversal are set to 0, the excitatory reversal is set to 1, and leak conductance is fixed at 1. Each compartment voltage is therefore a conductance-weighted average. Two properties of this form matter for what follows: local sensitivities depend on both driving force $(E - V)$ and input resistance R^{tot} , and shunting inhibition corresponds to adding conductance when $E_{\text{inh}} \approx 0$.

2.1 Voltage Equation and Local Sensitivities

Consider compartment n with synaptic inputs j (activity x_j , reversal E_j , conductance $g_j^{\text{syn}} \geq 0$) and dendritic inputs from children (voltage V_j , conductance $g_j^{\text{den}} \geq 0$). The steady-state voltage is:

$$V_n = \frac{\sum_j E_j x_j g_j^{\text{syn}} + \sum_j V_j g_j^{\text{den}}}{\underbrace{\sum_j x_j g_j^{\text{syn}} + \sum_j g_j^{\text{den}} + 1}_{g_n^{\text{tot}}}}, \quad R_n^{\text{tot}} = 1/g_n^{\text{tot}}. \quad (1)$$

60 V_n is a convex combination of reversal potentials, child voltages, and leak. Writing $\mathcal{S}_n = \{0\} \cup$
61 $\{E_j\}_j \cup \{V_j\}_j$, we have $\min \mathcal{S}_n \leq V_n \leq \max \mathcal{S}_n$ and $0 < R_n^{\text{tot}} \leq 1$. The local sensitivities follow
62 directly:

Proposition 1 (Local Sensitivities).

$$\frac{\partial V_n}{\partial g_i^{\text{syn}}} = x_i R_n^{\text{tot}} (E_i - V_n), \quad \frac{\partial V_n}{\partial V_i} = g_i^{\text{den}} R_n^{\text{tot}}, \quad \frac{\partial V_n}{\partial g_i^{\text{den}}} = R_n^{\text{tot}} (V_i - V_n). \quad (2)$$

63 2.2 Shunting Inhibition as Divisive Gain Control

64 An inhibitory synapse with $E_{\text{inh}} \approx 0$ contributes current $(0 - V_n)x_j g_j^{\text{syn}}$ and increases g_n^{tot} . Its
65 sensitivity is $\partial V_n / \partial g_j^{\text{syn}} = -x_j R_n^{\text{tot}} V_n$, which corresponds to multiplicative attenuation (divisive
66 normalization). Shunting is divisive at the voltage level, but its effect on firing rates can be subtractive
67 in some regimes [7]. Inhibitory plasticity can balance excitation dynamically [8]; our learned
68 inhibitory conductances serve an analogous role.

69 2.3 Exact Gradients for Dendritic Trees

70 Let V_0 be the somatic output, $\hat{y} = W_{\text{dec}} V_0$ the decoder, and $\delta_0 = W_{\text{dec}}^\top (\partial L / \partial \hat{y})$ the somatic error.

71 **Theorem 1** (Conductance-Stage Backpropagation on a Dendritic Tree). *For a rooted dendritic tree*
72 *with soma at node 0 and unique parent $p(n)$ for each non-somatic compartment, if each branch*
73 *passes its pre-activation voltage directly to its parent, then the loss gradient satisfies*

$$\frac{\partial L}{\partial V_n} = \frac{\partial L}{\partial V_{p(n)}} R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}, \quad (3)$$

74 *with boundary condition $\partial L / \partial V_0 = \delta_0$. Because the graph is a tree, each compartment has a unique*
75 *path to the soma. Defining the path gain*

$$\alpha_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightsquigarrow 0)} R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad \frac{\partial L}{\partial V_n} = \alpha_n \delta_0, \quad (4)$$

76 *with $\alpha_0 = 1$.*

77 *Proof.* Apply the chain rule on the tree-structured computation graph using Prop. 1. □

78 **Corollary 1** (Local–Global Factorization). *The exact synaptic gradient at compartment n factorizes*
79 *as:*

$$\frac{\partial L}{\partial g_i^{\text{syn}}} = \underbrace{x_i R_n^{\text{tot}} (E_i - V_n)}_{\text{synapse-local eligibility}} \cdot \underbrace{\frac{\partial L}{\partial V_n}}_{\text{compartment error}}, \quad (5)$$

80 *where the eligibility term depends only on quantities available at synapse i (presynaptic activity*
81 *x_i , input resistance R_n^{tot} , driving force $E_i - V_n$), and the compartment error $\partial L / \partial V_n$ is the sole*
82 *non-local quantity.*

83 Under identity upward transfer, this factorization isolates the only non-local quantity as $\partial L / \partial V_n =$
84 $\alpha_n \delta_0$. With a post-voltage activation $a_n = f_n(V_n)$ transmitted to the parent, the exact recursion
85 becomes $\partial L / \partial V_n = f'_n(V_n) (\partial L / \partial V_{p(n)}) R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}$, so the compartment error remains the
86 sole non-local term but the effective path gain acquires local activation-derivative factors. Any
87 practical local rule therefore depends entirely on how well a broadcast signal approximates that exact
88 compartment error, which we evaluate in Sec. 4.2. In the models studied here, shunting improves
89 this approximation by reducing and stabilizing R_n^{tot} , which narrows the spread of local sensitivities
90 across the tree. The theorem is stated for the pre-activation compartment voltage V_n produced by
91 the conductance stage itself. In the empirical networks, each dendritic branch layer may optionally
92 apply a post-voltage reactivation $f(V_n)$ before passing activity to the next stage; unless otherwise
93 noted, the manuscript sweeps use a learnable bounded `param_tanh` reactivation. Our exact-error
94 diagnostics are computed on the hooked pre-activation voltage, so the factorization check and
95 compartment-error analyses target the conductance-stage quantity defined above rather than the
96 post-activation activity.

3 Local Learning Rules

3.1 Broadcast Error Approximation

We approximate the exact compartment error $\partial L / \partial V_n$ (Corollary 1) with a broadcast signal e_n derived from the somatic/core error δ_0 . For a linear readout this reduces to $\delta_0 = W_{\text{dec}}^\top (\partial L / \partial \hat{y})$; for a nonlinear decoder we use the corresponding decoder-input Jacobian product $J_{\text{dec}}(h_{\text{core}})^\top (\partial L / \partial \hat{y})$ so that the broadcast remains defined in the same core-space coordinates. We consider five broadcast modes of increasing structure:

- **Scalar:** $e_n = \bar{\delta} \cdot \mathbf{1}$, where $\bar{\delta} = \text{mean}(\delta_0)$ is a single scalar broadcast to all compartments.
- **Per-soma:** $e_n = \delta_0$ when the layer dimension matches the output; layers with mismatched dimensions fall back to scalar.
- **Low-rank:** $e_n = \Gamma_K(\delta_0)$, where Γ_K uses a small number K of random broadcast channels to project the somatic error into a layer-specific vector field.
- **Local mismatch:** $e_n = (1-\alpha)\bar{\delta} \cdot \tilde{m}_n + \alpha\bar{\delta}$, where $\tilde{m}_n$ is the RMS-normalized, batch-centered parent-child voltage difference $P_n - V_n$ and $\alpha = 0.2$.
- **Pathway-vector:** $e_n = \alpha\bar{\delta} \cdot \mathbf{1} + (1-\alpha)\Gamma_n(\delta_0)$, where Γ_n projects the somatic error into router-inferred pathway roles, gates those roles by local pathway activity, and transports the resulting vector hierarchically through the block-structured dendritic tree.

We use per-soma broadcast by default. In our experiments, the local-mismatch variant performs much worse (Appendix A), indicating that broadcast quality is critical. In the headline feedforward architectures, however, hidden widths exceed decoder output dimension, so this default per-soma mode falls back to a shared scalar/rank-1 field (Appendix Table S13). Under shunting, that deliberately low-bandwidth broadcast is often sufficient on standard classification tasks, while the cue-routing benchmark in Sec. 4.9 is used to compare rank-1, random low-rank, and pathway-structured feedback when different branches should receive different teaching signals. Appendix Table S11 summarizes the broader set of implemented model families, training strategies, and broadcast modes used throughout the study.

3.2 Three-Factor Rule (3F)

Definition 1 (3F Update). *For synaptic and dendritic conductances:*

$$\Delta g_j^{\text{syn}} = \eta \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B, \quad \Delta g_j^{\text{den}} = \eta \langle R_n^{\text{tot}} (V_j - V_n) e_n \rangle_B, \quad (6)$$

where $\langle \cdot \rangle_B$ denotes the batch average.

The three factors are presynaptic activity x_j (or a voltage difference), postsynaptic modulation through driving force and input resistance, and the broadcast error e_n . The same rule applies to excitatory and inhibitory synapses. The sign difference comes from the driving force $(E_j - V_n)$.

Additive control. Rule (6) is the local gradient of the *shunting* voltage $V_n = \sum E_j g_j x_j / g_n^{\text{tot}}$. For the additive control ($V_n = \sum g_j x_j$, no divisive normalization), the correct local gradient is simpler: $\Delta g_j^{\text{syn}} = \eta \langle x_j e_n \rangle_B$, with no driving-force or R_n^{tot} terms ($R_n^{\text{tot}} \equiv 1$ by definition since there is no denominator). Throughout, each architecture uses the learning rule derived from its own forward-pass dynamics. This keeps the comparison architecture-matched rather than applying a shunting-derived rule to an additive model.

3.3 Practical Heuristic Wrappers: 4F and 5F

4F (heuristic attenuation proxy). In the conductance-stage recursion (4), the path gain α_n attenuates the somatic error differently at each compartment. To compensate without explicitly computing that gain, 4F uses an online correlation proxy

$$\rho_n = \text{Cov}(\bar{V}_n, \bar{V}_0) / (\sqrt{\text{Var}(\bar{V}_n) \text{Var}(\bar{V}_0)} + \varepsilon).$$

High ρ_n up-weights branches whose voltages covary with the soma; low ρ_n down-weights branches where the broadcast is likely to be a poor proxy.

141 **5F (confidence-weighted 4F).** 5F adds a second heuristic factor

$$\phi_n = \text{Var}(V_n) / (\sigma_{\text{res}}^2 + \varepsilon),$$

142 where σ_{res}^2 is the residual variance of V_n after linear regression on the parent voltage. Operationally,
 143 ϕ_n is a branch-wise confidence gate: it becomes large when local voltages are strong and stable
 144 relative to unexplained residual fluctuations, and small when local fluctuations dominate. We clamp it
 145 to $[0.25, 4.0]$ for stability. This factor is empirical, not theorem-derived. The paper’s core claim rests
 146 on path-gain concentration and the oracle experiment; the gain from ϕ_n should be read as practical
 147 denoising rather than theoretical necessity.

Proposition 2 (5F Update).

$$\Delta g_j^{\text{syn}} = \eta \rho_n \phi_n \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B. \quad (7)$$

148 A feedback-alignment-style random-broadcast argument is included in Appendix C; it is supportive
 149 but not central to the main mechanism claim.

150 3.4 Structured pathway feedback and branch-role plasticity

151 For tasks with latent pathway structure, scalar broadcast can erase the distinction between different
 152 dendritic branches that should be credited differently. Recent experimental evidence supports the
 153 existence of vectorized instructive signals in cortical dendrites [43], motivating the structured feedback
 154 variant we introduce here. We therefore augment the local rule with a structured cue-routing variant,
 155 *PV-LocalCA*.

156 **Hierarchical pathway-vector broadcast.** The encoder/router defines a low-dimensional role basis
 157 over latent pathways. Instead of collapsing δ_0 to a scalar or per-soma vector, we project it into this
 158 role basis and gate each role by local pathway activity:

$$e_n^{\text{PV}} = \sum_{k=1}^K q_{n,k} c_k,$$

159 where $c_k = \langle r_k, \delta_0 \rangle$ are role coefficients and $q_{n,k}$ are local branch-role weights. Scalar broadcast is
 160 the special case $K=1$, while *PV-LocalCA* is a structured rank- K extension rather than a separate
 161 learning principle.

162 **Router-derived branch roles.** Rather than assigning hand-coded apical/basal labels, each branch
 163 j receives a normalized role profile r_j inferred from router assignments and incoming block weights.
 164 Plasticity is then gated by branch selectivity and by a synapse-specific role-alignment factor $\langle r_i, r_j \rangle$,
 165 biasing learning toward pathway-consistent afferents without imposing a fixed branch taxonomy.

166 4 Experiments

167 4.1 Setup

168 We evaluate two settings: a *capacity-calibrated* regime, where matched architectures achieve strong
 169 backprop performance, and *controlled sweeps* that isolate inhibition strength, broadcast mode, and
 170 architecture. Our main datasets are MNIST, Fashion-MNIST [37], and three synthetic tasks: *context*
 171 *gating*, *noise resilience*, and *cue integration*; formal definitions are in Appendix I. An older *info*
 172 *shunting* benchmark remains in the codebase as a signed-input exploratory stress test, but it is no
 173 longer part of the headline corrected result set because it violates the nonnegative-drive assumption
 174 of the conductance model. Architectures include point MLP baselines and dendritic cores with either
 175 additive integration or shunting. Main comparisons are architecture-matched additive vs. shunting
 176 cores and local vs. backprop training; DFA appears only as a supplementary baseline. Most headline
 177 MNIST, Fashion-MNIST, and context-gating results use 5F with per-soma broadcast. Because
 178 the hidden widths in these feedforward settings exceed the decoder output dimension, that default
 179 per-soma implementation reduces to a shared scalar/rank-1 broadcast in the main mechanism and
 180 competence sweeps; the higher-rank controls then use explicit low-rank, path-propagation, path-
 181 transport, or pathway-vector variants. Unless otherwise stated, dendritic layers also use the learnable

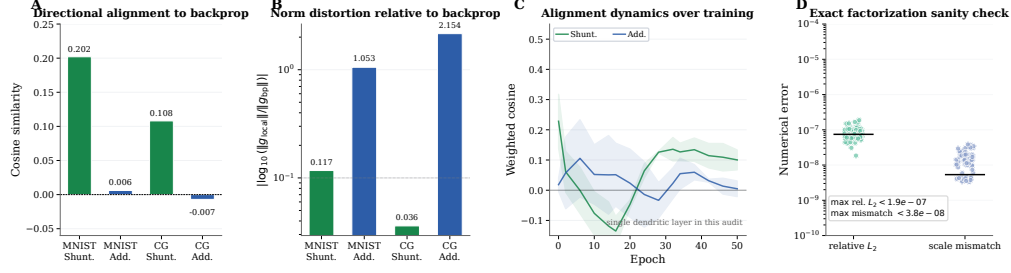


Figure 2: **Gradient-fidelity diagnostic.** (A) Weighted cosine similarity between local and backprop gradients: shunting (green) is consistently more aligned than additive (blue) on both MNIST and context gating. (B) Gradient scale mismatch on the same matched-weight comparisons. Shunting is not only better aligned, but also substantially closer in norm. (C) Per-layer cosine similarity over training epochs. Shunting proximal layers approach high positive alignment; additive layers remain near zero. (D) Exact-factorization sanity check across all corrected theory-diagnostic conditions, reported with relative L_2 reconstruction error and scale mismatch. Reconstructed gradients match autograd to numerical precision, showing that LocalCA’s only approximation is in the broadcast substitute for the compartment error.

182 post-voltage reactivation `param_tanh`; Appendix Tables S8 and S9 then audit activation shape in a
 183 more controlled setting by comparing identity (none) against a frozen `param_tanh` under matched
 184 additive/shunting initialization. All headline dendritic runs keep synaptic and dendritic conductances
 185 nonnegative via positive parameter transforms and enforce a nonnegative first-layer synaptic drive.
 186 Concretely, image-based inputs are used in $[0, 1]$, the corrected noise-resilience benchmark clips
 187 projected Gaussian corruption back to $[0, 1]$, context gating uses a nonnegative contextual figure-
 188 ground MNIST construction, and cue integration uses one-hot context plus cue values clipped to $[0, 1]$.
 189 The cue-routing benchmark is used to compare rank-1, random low-rank, and pathway-structured
 190 feedback in a routed setting, and the inhibition sweeps additionally use a transported-error oracle
 191 derived from Theorem 1 as an upper bound rather than as the proposed biological rule. Headline
 192 mechanistic, oracle, competence, activation-audit, corrected routed, and CIFAR mechanism-extension
 193 results use 5 seeds per condition; a smaller set of appendix-only support sweeps still use 3 seeds,
 194 while 1–2 seed screens are treated as exploratory and are either omitted from the main claims or
 195 explicitly labeled as such in the appendix.

196 4.2 From Exact Factorization to Credit-Signal Fidelity

197 To test whether these performance differences correspond to better credit signals, we compare
 198 local and backprop gradients on the same batch at the same parameter values. For each parameter
 199 tensor p , we compute directional alignment (cosine similarity) and scale mismatch, defined as
 200 $|\log_{10}(\|g_p^{\text{local}}\|/\|g_p^{\text{bp}}\|)|$, aggregated by parameter count.

201 Across matched-weight comparisons, shunting is consistently more aligned with backprop and
 202 substantially closer in norm than the additive control: on MNIST, weighted cosine rises from 0.006
 203 to 0.202 and scale mismatch falls from 1.053 to 0.117; on context gating, cosine rises from -0.007
 204 to 0.108 and scale mismatch falls from 2.154 to 0.036. Figure 2C shows that this advantage appears
 205 throughout training, especially in proximal layers and dendritic conductances. Figure 2D confirms
 206 that the theory is operational in code under the corrected positive-input setup: reconstructing gradients
 207 from Corollary 1 using exact compartment errors yields weighted relative reconstruction error below
 208 2×10^{-7} and weighted scale mismatch below 4×10^{-8} across runs. In a few low-norm parameter
 209 blocks, cosine becomes numerically unstable even though the reconstruction itself is essentially exact,
 210 so we report the norm-based diagnostics in the figure. The only approximation made by LocalCA is
 211 therefore the broadcast substitute for $\partial L/\partial V_n$.

212 4.3 Mechanistic Chain: Path Gains, Broadcast Fidelity, and Oracle Transport

213 Figure 3 provides the paper’s central empirical chain: shunting concentrates conductance-stage path
 214 gains, which improves direct compartment-error fidelity, which in turn makes low-bandwidth local

learning effective; the transported-error oracle then isolates how much of the remaining gap is due to broadcast quality rather than other architectural differences.

Panel A measures the coefficient of variation of the exact conductance-stage path gains α_n across compartments. On MNIST, shunting keeps this distribution tightly concentrated across inhibition levels (CV about 0.21–0.43), whereas the additive control remains much broader (CV about 0.90–1.16). Panel B then measures the sole non-local term from Corollary 1, $\partial L/\partial V_n$, directly. On the corrected noise-resilience sweep, per-soma broadcast tracks the exact compartment error somewhat better under shunting than under additive integration once inhibition is present: additive remains positively aligned throughout (about 0.14–0.19 across the sweep), while shunting rises to about 0.20–0.24 in the inhibited regimes after starting from a comparable low-inhibition baseline. The transported-error oracle obtained by recursively applying those conductance-stage path gains improves fidelity sharply in both architectures, reaching about 0.95–0.99 for additive and essentially 1.0 for shunting once inhibition is present, showing that the theorem defines an operational upper bound on low-bandwidth broadcast.

Panel C trains the same local rule using that oracle signal. On noise resilience, additive learning rises from 66.8% at $N_I=0$ to about 81–84% across the inhibited regimes under per-soma broadcast, and transported broadcast lifts those same conditions to about 92–94%. Shunting rises from 54.3% at $N_I=0$ to about 81–86% under per-soma and to about 94.3–95.2% under transported broadcast. On MNIST, transported broadcast reaches about 95.4–96.6% for additive and 96.5–97.3% for shunting. The CIFAR-10 inset shows the same harder-data pattern in the corrected compact E/I family: shunting rises from 28.8% under per-soma broadcast to 47.4% under transported error, nearly matching its 49.5% backprop ceiling. These oracle results strongly support path-gain fidelity as the dominant bottleneck for LocalCA performance, while the smaller remaining additive–shunting gap under matched transported broadcast suggests an additional contribution from conductance-shaped local sensitivities.

Panel D shows that coarse broadcast can still be enough once local sensitivities are well scaled: 4-bit quantization preserves full performance (61.6% vs. 61.7%; $p > 0.9$), while sign-only (37.5%) and sparse top-30% (38.2%) degrade sharply. The corrected rank-bridge sweep in Appendix Table S7 adds one constructive intermediate control. On the same shunting noise-resilience regime, per-soma reaches 0.662 ± 0.016 , the current path-propagation proxy rises to 0.762 ± 0.047 , random low-rank broadcast reaches 0.764 ± 0.025 at $K=4$, and exact path transport reaches 0.847 ± 0.018 . So both morphology-aware propagation and higher-rank unstructured feedback now close a substantial fraction of the transported-oracle gap, although exact transported error remains clearly better. Normalization-only additive controls and DFA baselines (Appendix Figs. S7, S6) improve over weaker baselines without matching the shunting conductance model. Appendix Table S8 shows that this mechanism should still be interpreted in a specific activation operating regime rather than as an activation-agnostic claim.

4.4 Capacity-Calibrated Competence

Having established the mechanism, we next ask whether the same conditions improve end-task learning in architectures that are already competent under backpropagation. In the capacity-calibrated regime, shunting dendritic cores remain reasonably competitive under local learning: on MNIST, Fashion-MNIST, and context gating, the best local configuration (5F, per-soma broadcast, local decoder) stays within about 5–6 percentage points of the matched backprop ceiling (Table S4; Fig. 4). Within the local-rule family, 5F is the strongest practical wrapper, but it remains heuristic rather than theorem-derived; Appendix Tables S1, S3, and S2 summarize that division of labor across rules and broadcast choices.

4.5 Regime Dependence of the Shunting Advantage

The benefit of shunting varies across tasks. On MNIST with matched per-soma broadcast, shunting outperforms additive by about 2 percentage points; on the corrected nonnegative-input noise-resilience benchmark, the gap is much larger, reaching about +50.3 pp at $N_I=10$ inhibitory synapses per branch. Figure 4 shows the full dose-response: the shunting advantage grows with inhibitory conductance strength on that inhibition-sensitive task, whereas additive cores remain more fragile in the same regime.

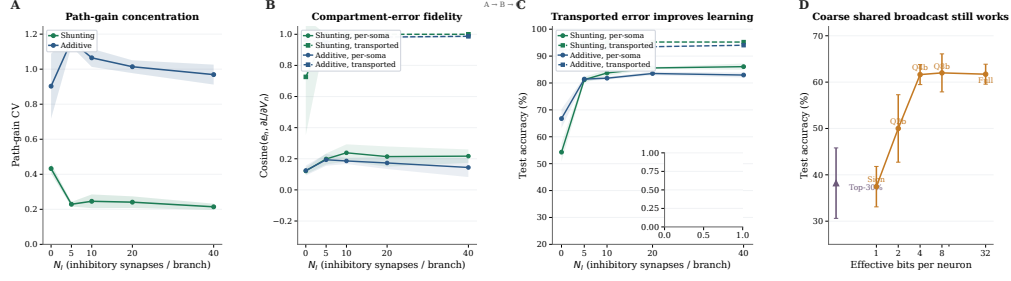


Figure 3: **Mechanistic evidence.** (A) Conductance-stage path-gain dispersion on MNIST, measured as the coefficient of variation of the exact tree attenuation α_n across compartments. Shunting keeps these conductance-stage path gains much more concentrated than the additive control as inhibition increases. (B) Direct compartment-error fidelity on noise resilience. Solid lines compare the per-soma broadcast e_n to the exact compartment error $\partial L/\partial V_n$; dashed lines show the transported-error oracle obtained by recursively applying conductance-stage path gains. Under the corrected positive-input setup, shunting still improves the simple per-soma broadcast in the inhibited regimes, while the transported error oracle defines the remaining approximation gap and becomes nearly exact once inhibition is present. (C) Learning with the transported-error oracle on noise resilience. Improving the broadcast field is enough to recover near-ceiling local learning, especially in the additive architecture where per-soma broadcast fails most severely; the CIFAR-10 inset shows the same harder-data ladder for shunting in the corrected compact E/I family. (D) Broadcast bandwidth. Four-bit quantization preserves full performance; only sign-only and sparse modes degrade sharply.

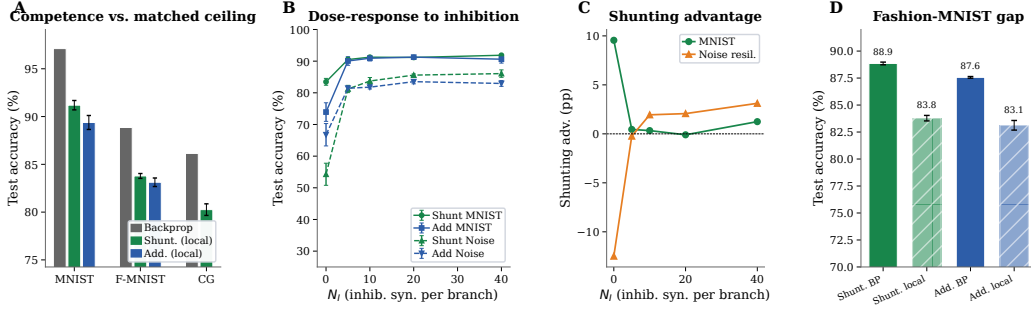


Figure 4: **Capacity-calibrated competence and regime dependence.** (A) Backprop ceiling (gray) vs. best local rule (5F per-soma) for shunting (green) and additive (blue) cores on MNIST, Fashion-MNIST, and context gating, using the refreshed 5-seed ceilings. (B) N_I dose-response: test accuracy vs. inhibitory synapses per branch (N_I) for shunting and additive cores on MNIST (solid) and noise resilience (dashed), updated from the corrected nonnegative-input sweep. (C) Shunting advantage (Δ , in percentage points) vs. N_I . (D) Fashion-MNIST standard-vs-local comparison from the corrected five-seed sweep, showing a smaller but still consistent shunting advantage on the cleaner vision benchmark.

267 4.6 Morphology Shifts the Inhibitory Operating Regime

268 The inhibitory regime in which shunting helps most is not fixed across dendritic trees. Figure 5 shows
 269 the corrected morphology $\times$ inhibition sweep on noise resilience. The main pattern is not monotonic
 270 “more inhibition is better,” but rather a morphology-dependent ridge: depth-2 trees show their largest
 271 shunting gains near $N_I=0$, whereas deeper depth-3 trees retain their strongest gains around $N_I=5$.
 272 This is consistent with the broader mechanistic picture that shunting helps by shaping the spread of
 273 effective compartment sensitivities, and suggests that tree geometry changes where that operating
 274 point lies.

275 4.7 Additional Stress Tests

276 Across three additional stress tests — increased dendritic depth, noisy broadcast signals, and Fashion-
 277 MNIST — the same qualitative pattern persists: shunting degrades more gracefully than additive,

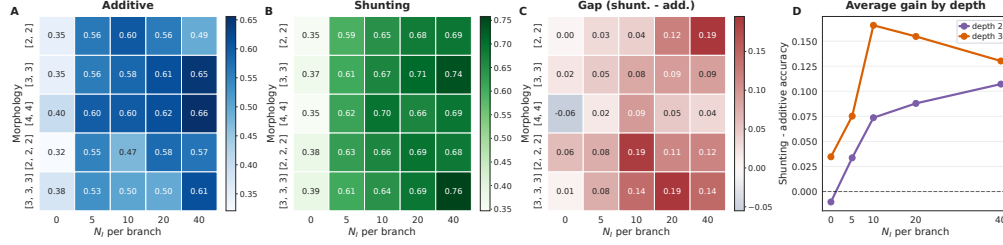


Figure 5: **Morphology shifts the inhibitory regime where shunting helps most.** (A) Additive LocalCA remains comparatively flat across morphologies and inhibitory synapse counts, with only isolated high-variance pockets. (B) Shunting LocalCA shows a clear morphology-dependent regime: the strongest performance appears at low or moderate inhibition, and the best N_I shifts with tree shape. (C) The shunting-minus-additive gap is largest for wider two-level trees at low inhibition and for deeper trees at modest inhibition, then weakens or vanishes at high inhibition. (D) Averaging by depth highlights the same crossover: depth-2 trees peak near $N_I=0$, whereas depth-3 trees retain their largest shunting advantage around $N_I=5$.

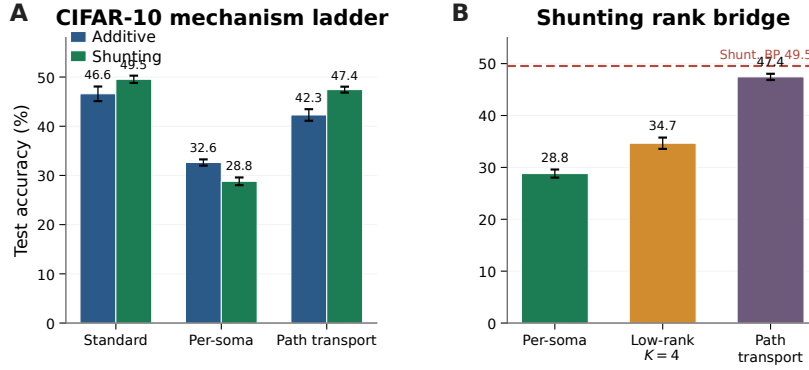


Figure 6: **Corrected CIFAR-10 mechanism extension in the stronger compact E/I family.** (A) Matched additive vs. shunting ladder with a nonlinear decoder and decoder-aware soma-error mapping. In this stronger family, shunting is competitive under standard training ($49.5 \pm 0.7\%$) and transported-error LocalCA nearly recovers that ceiling ($47.4 \pm 0.6\%$), while naive rank-1 per-soma remains much weaker. (B) Shunting rank bridge on the same family. Random low-rank feedback improves over per-soma, but exact transported error remains the strongest signal.

while the largest gains still appear on tasks that stress the credit signal most strongly. We keep the full curves in Appendix Fig. S5 because they support, rather than define, the core mechanistic claim.

4.8 Harder-Data Check: Corrected CIFAR-10 Mechanism Extension

To make sure the mechanism is not an artifact of near-saturated MNIST-style regimes, we repeated the core ladder on flattened CIFAR-10 in the stronger compact explicit-E/I family where shunting is already competitive under backpropagation. Figure 6 shows the corrected 5-seed result after fixing LocalCA's soma-error mapping for nonlinear decoders. In this family, shunting is stronger than additive under standard training ($49.5 \pm 0.7\%$ vs. $46.6 \pm 1.5\%$), and transported-error LocalCA again behaves as the clean upper bound: additive rises from $32.6 \pm 0.6\%$ under rank-1 per-soma broadcast to $42.3 \pm 1.2\%$ under transported error, while shunting rises from $28.8 \pm 0.8\%$ under per-soma to $34.7 \pm 1.1\%$ with random low-rank $K=4$ and $47.4 \pm 0.6\%$ with transported error. So the harder-data result is not that shunting always wins under naive local broadcast, but that the same broadcast-fidelity ladder still governs performance once the architecture and decoder mapping are handled fairly.

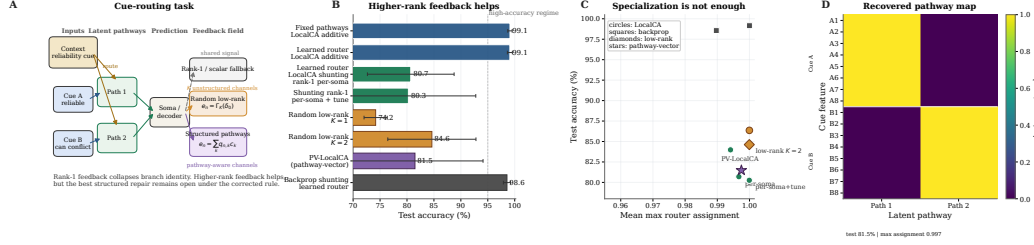


Figure 7: **Structured feedback when rank-1 broadcast fails.** (A) Cue-routing task schematic. Context determines which noisy cue is reliable on each trial, so different dendritic pathways should receive different teaching signals. (B) Seed-averaged diagnosis and focused tuning (5 seeds per condition). Learned-routing additive local learning remains high, whereas learned-routing shunting variants remain substantially less stable under the corrected rule; the panel compares rank-1, random low-rank, and pathway-structured variants directly. (C) Test accuracy vs. router specialization across learned-router conditions. High specialization alone does not guarantee success. (D) Recovered pathway assignments for a representative learned-router run. Cue-A and cue-B features segregate cleanly into separate latent pathways even when feedback quality, rather than router discreteness, limits performance.

4.9 Extension: Structured Feedback When Per-Soma Broadcast Fails

The main paper focuses on low-bandwidth scalar or per-soma broadcast. We therefore include a harder cue-integration benchmark only as an exploratory extension: here different dendritic pathways should receive different teaching signals, so rank-1 feedback is insufficient by construction. Figure 7 shows that the benchmark itself is learnable, but learned-routing shunting LocalCA remains the hard case. Under the corrected activation-aware rule, default rank-1 per-soma broadcast stays unstable at roughly 0.81 ± 0.13 test accuracy; moving beyond rank-1 helps, with random low-rank feedback rising from 0.742 ± 0.022 at $K=1$ to 0.846 ± 0.082 at $K=2$, while the current tuned pathway-vector variant reaches 0.815 ± 0.126 . The robust conclusion is therefore the failure mode: once rank-1 feedback collapses pathway identity, higher-rank feedback can help, but the best structured repair remains open.

5 Related Work

High-performance local learning. Recent methods such as AugLocal, CCL, and DLL have substantially raised the benchmark-performance bar for local learning on standard architectures [33, 34, 36]. Our aim is therefore not to claim the strongest benchmark accuracy among local-learning methods, but to identify the biophysical mechanism that determines when low-bandwidth credit signals become useful in conductance-based dendritic networks.

Local credit-assignment dynamics. Random feedback, DFA, DTP, DFC, equilibrium-style methods, and predictive-coding approaches solve locality through different feedback or inference dynamics [9, 10, 13, 15, 16, 25–29, 32]. EBD [35] is especially close in spirit because it studies when direct error broadcast can be made useful by shaping the broadcast field itself. Our contribution is complementary: rather than changing the broadcast objective or inference dynamics, we identify a conductance-based inductive bias—conductance-stage path-gain concentration under shunting—that makes even a simple low-bandwidth broadcast more faithful to the exact compartment error.

Dendrites for robustness and context. Dendritic trees support nonlinear computation [3, 4, 30] and have inspired learning schemes based on segregated dendrites, dendritic prediction errors, burst-dependent plasticity, and latent equilibrium [5, 11, 12, 22–24]. Outside supervised iid classification, active-dendrite architectures can also improve robustness in dynamic multi-task settings where point-MLP baselines suffer interference [44]. Our cue-routing extension asks a related question—when dendritic structure helps preserve pathway-specific computation—but uses conductance-based normalization and structured credit signals rather than sparse kWTA gating.

Neuroscience grounding. Shunting implements divisive normalization [6], though its rate effects can be subtractive in some regimes [7]; inhibitory conductance also modulates gain and signal-to-noise [31]. Recent evidence for vectorized instructive signals in cortical dendrites [43] further supports the structured-feedback extension studied here. To our knowledge, prior work has not tied exact conductance-stage path gains, direct gradient-fidelity diagnostics, and shunting inhibition into one causal account of when local dendritic learning works.

6 Discussion

We started from exact gradients for a conductance-based dendritic tree and asked when a somatic broadcast can serve as a useful proxy for compartment-specific errors. In the models studied here, the answer depends strongly on the inhibitory regime: shunting changes the conductance denominator, stabilizes effective local sensitivities, improves agreement between local and backprop gradients, and yields better local learning than an additive control, especially in noisy or inhibition-dependent settings. The morphology $\times$ inhibition map in Fig. 5 suggests that tree shape shifts this operating regime, and explicit normalization recovers only part of the benefit. This is therefore a mechanism paper, not a benchmark paper. The activation audit in the appendix shows that the mechanism lives in the bounded `param_tanh` regime used throughout the main experiments, and the paired empirical/theory activation tables show why: freezing that activation often broadens path-gain dispersion and weakens per-soma fidelity in the corrected positive-input setting. The oracle/rank-bridge controls then sharpen the remaining open problem: finding a strictly local morphology-aware approximation that closes more of the transported-error gap. All headline experiments reported here now also respect the basic conductance assumptions directly: synaptic and dendritic weights remain nonnegative throughout training, and the first synaptic drive is kept nonnegative by the transfer stage and by dataset preprocessing. The corrected flattened-CIFAR-10 extension in Fig. 6 supports the same diagnosis in a stronger compact E/I family: once the local rule maps decoder error back to decoder-input/soma space, transported error again acts as a meaningful upper bound and shunting remains competitive on harder data.

Limitations. Our best accuracy (91.4% MNIST, 83.8% Fashion-MNIST) remains below several recent local-learning methods that operate in abstract compartments or standard activation spaces [33–36]. This reflects the constraints of conductance-based voltage space: bounded voltages, nonnegative conductances, nonnegative presynaptic drive, and a denominator-heavy computation graph. The corrected flattened-CIFAR-10 extension in Fig. 6 and Table S6 is a harder stress test rather than a claim of benchmark leadership. There shunting is competitive or better than additive under standard training and under transported-error LocalCA, but naive rank-1 per-soma broadcast still leaves a substantial gap, indicating that harder visual tasks demand better broadcast quality and likely richer front ends than flattened inputs alone. More broadly, naive local broadcasts remain much weaker than per-soma on MNIST, structured routed settings remain open, per-soma broadcast is still neuron-global, depth remains challenging, shunting is not gain-invariant in practice, and the FA and weight-statistics appendices are supportive rather than central evidence. We also do not claim the same empirical story under externally signed inputs; those legacy stress tests remain outside the corrected headline result set.

Broader relevance. The results are relevant to neuroscience, machine learning, and neuromorphic hardware. For neuroscience, they suggest a connection between divisive normalization and local credit fidelity. For machine learning, they show that conductance-based inductive biases can shape gradient geometry in ways that benefit local learning. For neuromorphic implementations, the update rules remain strictly local and parallelizable.

Testable predictions. Our framework predicts that (1) stronger perisomatic inhibition yields more precise synaptic plasticity; (2) GABA_A blockade selectively impairs multi-layer credit tasks; (3) the dendritic–somatic voltage correlation (ρ_n) increases during learning; and (4) learned excitatory weight distributions follow log-normal statistics whose width depends systematically on inhibitory conductance (Appendix G).

References

- [1] Koch, C. (1999). *Biophysics of Computation*. Oxford University Press.
- [2] Dayan, P., & Abbott, L. F. (2001). *Theoretical Neuroscience*. MIT Press.
- [3] Poirazi, P., Brannon, T., & Mel, B. W. (2003). Pyramidal neuron as two-layer neural network. *Neuron*, 37(6), 989–999.
- [4] London, M., & Häusser, M. (2005). Dendritic computation. *Annual Review of Neuroscience*, 28, 503–532.
- [5] Urbanczik, R., & Senn, W. (2014). Learning by the dendritic prediction of somatic spiking. *Neuron*, 81(3), 521–528.
- [6] Carandini, M., & Heeger, D. J. (2012). Normalization as a canonical neural computation. *Nature Reviews Neuroscience*, 13(1), 51–62.
- [7] Holt, G. R., & Koch, C. (1997). Shunting inhibition does not have a divisive effect on firing rates. *Neural Computation*, 9(5), 1001–1013.
- [8] Vogels, T. P., Sprekeler, H., Zenke, F., Clopath, C., & Gerstner, W. (2011). Inhibitory plasticity balances excitation and inhibition in sensory pathways and memory networks. *Science*, 334(6062), 1569–1573.
- [9] Lillicrap, T. P., Cownden, D., Tweed, D. B., & Akerman, C. J. (2016). Random synaptic feedback weights support error backpropagation for deep learning. *Nature Communications*, 7, 13276.
- [10] Nøkland, A. (2016). Direct feedback alignment provides learning in deep neural networks. *NeurIPS*, 29.
- [11] Guerguiev, J., Lillicrap, T. P., & Richards, B. A. (2017). Towards deep learning with segregated dendrites. *eLife*, 6, e22901.
- [12] Sacramento, J., et al. (2018). Dendritic cortical microcircuits approximate the backpropagation algorithm. *NeurIPS*, 31.
- [13] Whittington, J. C., & Bogacz, R. (2019). Theories of error back-propagation in the brain. *Trends in Cognitive Sciences*, 23(3), 235–250.
- [14] Richards, B. A., & Lillicrap, T. P. (2019). Dendritic solutions to the credit assignment problem. *Current Opinion in Neurobiology*, 54, 28–36.
- [15] Scellier, B., & Bengio, Y. (2017). Equilibrium propagation. *Frontiers in Computational Neuroscience*, 11, 24.
- [16] Song, Y., et al. (2024). Inferring neural activity before plasticity as a foundation for learning beyond backpropagation. *Nature Neuroscience*, 27, 348–358.
- [17] Gretton, A., et al. (2005). Measuring statistical dependence with Hilbert-Schmidt norms. *ALT*, 63–77.
- [18] Frémaux, N., & Gerstner, W. (2016). Neuromodulated spike-timing-dependent plasticity, and theory of three-factor learning rules. *Frontiers in Neural Circuits*, 9, 85.
- [19] Bellec, G., et al. (2020). A solution to the learning dilemma for recurrent networks of spiking neurons. *Nature Communications*, 11, 3625.
- [20] Welford, B. P. (1962). Note on a method for calculating corrected sums of squares and products. *Technometrics*, 4(3), 419–420.
- [21] Turrigiano, G. G. (2008). The self-tuning neuron: synaptic scaling of excitatory synapses. *Cell*, 135(3), 422–435.

- 415 [22] Payeur, A., et al. (2021). Burst-dependent synaptic plasticity can coordinate learning in hierar-
416 chical circuits. *Nature Neuroscience*, 24(7), 1010–1019.
- 417 [23] Greedy, W., et al. (2022). Single-phase deep learning in cortico-cortical networks. *NeurIPS*, 35.
- 418 [24] Haider, P., et al. (2021). Latent equilibrium. *NeurIPS*, 34.
- 419 [25] Hinton, G. (2022). The forward-forward algorithm. *arXiv:2212.13345*.
- 420 [26] Dellaferrera, G., & Kreiman, G. (2022). Error-driven input modulation: Solving the credit
421 assignment problem without a backward pass. *Proceedings of the 39th International Conference*
422 *on Machine Learning, Proceedings of Machine Learning Research*, 162, 4937–4955.
- 423 [27] Lee, D.-H., et al. (2015). Difference target propagation. *ECML*, 498–515.
- 424 [28] Meulemans, A., et al. (2021). Credit assignment in neural networks through deep feedback
425 control. *NeurIPS*, 34.
- 426 [29] Millidge, B., Seth, A. K., & Buckley, C. L. (2021). Predictive coding: A theoretical and
427 experimental review. *arXiv:2107.12979*.
- 428 [30] Koch, C., Poggio, T., & Torre, V. (1983). Nonlinear interactions in a dendritic tree. *PNAS*, 80(9),
429 2799–2802.
- 430 [31] Silver, R. A. (2010). Neuronal arithmetic. *Nature Reviews Neuroscience*, 11(7), 474–489.
- 431 [32] Max, K., Kriener, L., Pineda García, G., Nowotny, T., Jaras, I., Senn, W., & Petrovici, M.
432 A. (2024). Learning efficient backprojections across cortical hierarchies in real time. *Nature*
433 *Machine Intelligence*, 6(6), 619–630.
- 434 [33] Ma, C., Wu, J., Si, C., & Tan, K. C. (2024). Scaling supervised local learning with augmented
435 auxiliary networks. *International Conference on Learning Representations*.
- 436 [34] Lv, C., Xu, J., Lu, Y., Wang, X., Wang, Z., Xu, Z., Yu, D., Du, X., Zheng, X., & Huang, X.
437 (2025). Dendritic Localized Learning: Toward Biologically Plausible Algorithm. *Proceedings*
438 *of the 42nd International Conference on Machine Learning, Proceedings of Machine Learning*
439 *Research*, 267, 41682–41700.
- 440 [35] Erdogan, M., Pehlevan, C., & Erdogan, A. T. (2025). Error broadcast and decorrelation as a
441 potential artificial and natural learning mechanism. *NeurIPS 2025 Spotlight*, OpenReview.
- 442 [36] Kao, C.-H., & Hariharan, B. (2024). Counter-Current Learning: A biologically plausible dual
443 network approach for deep learning. *Advances in Neural Information Processing Systems*, 37.
- 444 [37] Xiao, H., Rasul, K., & Vollgraf, R. (2017). Fashion-MNIST: A novel image dataset for bench-
445 marking machine learning algorithms. *arXiv:1708.07747*.
- 446 [38] Song, S., Sjöström, P. J., Reigl, M., Nelson, S., & Chklovskii, D. B. (2005). Highly Nonrandom
447 Features of Synaptic Connectivity in Local Cortical Circuits. *PLoS Biology*, 3(3), e68.
- 448 [39] Buzsáki, G., & Mizuseki, K. (2014). The log-dynamic brain: How skewed distributions affect
449 network operations. *Nature Reviews Neuroscience*, 15(4), 264–278.
- 450 [40] The MICrONS Consortium et al. (2025). Functional connectomics spanning multiple areas of
451 mouse visual cortex. *Nature*, 640(8058), 435–447.
- 452 [41] Loewenstein, Y., Kuras, A., & Rumpel, S. (2011). Multiplicative Dynamics Underlie the
453 Emergence of the Log-Normal Distribution of Spine Sizes in the Neocortex In Vivo. *Journal of*
454 *Neuroscience*, 31(26), 9481–9488.
- 455 [42] Megías, M., et al. (2001). Total number and distribution of inhibitory and excitatory synapses
456 on hippocampal CA1 pyramidal cells. *Neuroscience*, 102(3), 527–540.
- 457 [43] Francioni, V., Tang, V. D., Toloza, E. H. S., Ding, Z., Brown, N. J., & Harnett, M. T. (2026).
458 Vectorized instructive signals in cortical dendrites. *Nature*. [https://doi.org/10.1038/s41586-026-](https://doi.org/10.1038/s41586-026-10190-7)
459 [10190-7](https://doi.org/10.1038/s41586-026-10190-7)

Dataset	Rule	Top-10 valid	Top-10 test
MNIST	3F	0.611	0.622
MNIST	4F	0.620	0.628
MNIST	5F	0.912	0.916
Context gating	3F	0.398	0.396
Context gating	4F	0.411	0.411
Context gating	5F	0.807	0.789

Table S1: **Rule-family ranking.** Top-10 mean across completed local-competence sweeps.

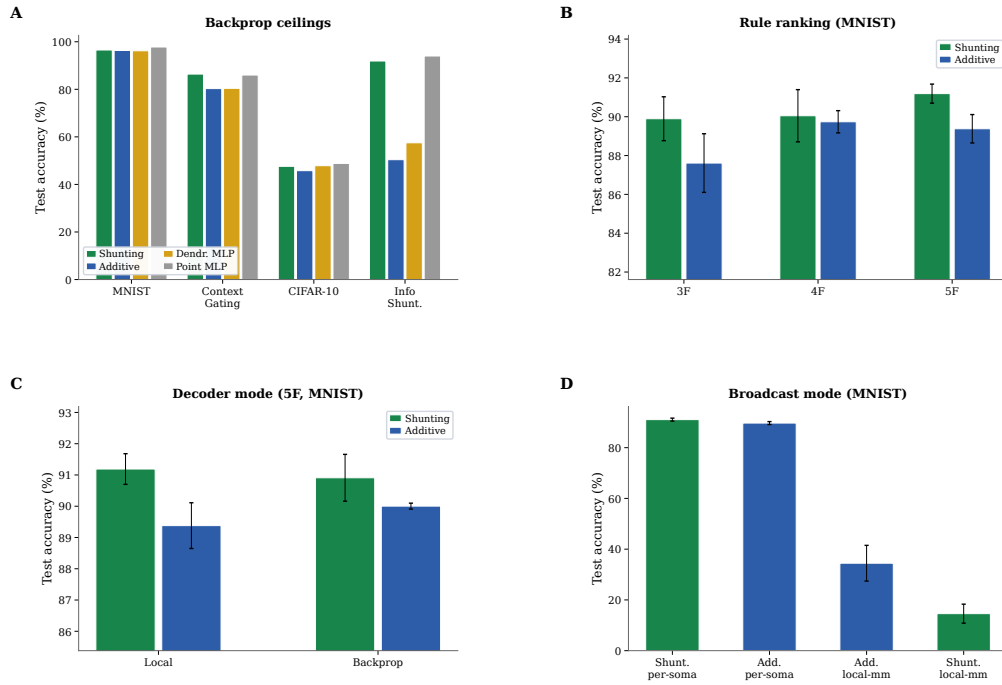


Figure S1: **Capacity calibration (supplementary).** (A) Phase 1 backprop ceilings across all architectures and datasets. (B) Rule-family ranking: 5F consistently outperforms 4F and 3F. (C) Decoder mode comparison (local vs. backprop decoder, 5F MNIST). (D) Broadcast mode comparison: per-soma is required for strong performance; local-mismatch fails.

- 460 [44] Iyer, A., Grewal, K., Velu, A., Souza, L. O., Forest, J., & Ahmad, S. (2022). Avoiding catastrophe: Active dendrites enable multi-task learning in dynamic environments. *Frontiers in*
461 *Neurobotics*, 16, 846219.
462

Component	Representative effect	Interpretation
Shunting architecture	In the corrected noise-resilience sweep, additive performs better at $N_I=0$ (0.668 vs. 0.543), but shunting overtakes it once inhibition is present (0.837 vs. 0.818 at $N_I=10$, 0.861 vs. 0.830 at $N_I=40$).	The shunting gain is operating-regime dependent rather than uniform: it appears once inhibition shapes conductance scaling strongly enough to help the shared broadcast.
Broadcast quality	On MNIST (5F), shunting per-soma reaches 0.912 ± 0.005 , while local-mismatch remains at 0.146 ± 0.046 .	Correctly matching the broadcast field to the somatic error matters far more than arbitrary local mismatch heuristics.
Transported oracle	Across inhibited noise-resilience regimes, additive local learning rises from roughly 81–84% under per-soma to 92–94% under transported broadcast; shunting rises from roughly 81–86% to 95–95.2%.	The oracle remains the strongest causal evidence that broadcast fidelity, not just architecture, is the dominant bottleneck.
Rule hierarchy	In the competence sweeps, MNIST improves from 0.622 (3F) and 0.628 (4F) to 0.916 (5F); context gating improves from 0.396/0.411 to 0.789.	5F is the strongest practical rule in this paper, though its extra factor remains heuristic rather than theorem-derived.
5F stability	Tightening the ϕ clamp to $[0.5, 2.0]$ changes MNIST by only about +0.4 pp; varying EMA from 0.05 to 0.20 changes it by less than 0.2 pp.	The practical gain from 5F is not especially brittle to its main tuning knobs.
Structured feedback	In the corrected cue-routing reruns, learned-routing additive local learning remains near ceiling while learned-routing shunting rises from 0.742 at random rank-1 to 0.846 at random rank-2; the current tuned pathway-vector variant reaches 0.815.	Routed tasks remain the main failure mode of rank-1 feedback, but under the corrected rule the main robust conclusion is that higher rank helps; the best structured fix is not yet settled enough to anchor a headline claim.
Random low-rank channels	On the corrected shunting noise-resilience bridge, random low-rank broadcast improves from 0.453 at $K=1$ to 0.764 at $K=4$ and 0.795 at $K=8$, while exact path transport reaches 0.847; on cue routing, the best corrected unstructured low-rank setting is $K=2$ at 0.846.	Higher-rank unstructured broadcast now closes a large fraction of the rank-1 gap, but it still does not recover the transported-oracle ceiling and does not by itself establish a structure-specific advantage.
Path propagation proxy	In the corrected shunting noise-resilience bridge, the current morphology-aware path-propagation proxy reaches 0.762 ± 0.047 , well above per-soma (0.662 ± 0.016) and close to random low-rank $K=4$ (0.764 ± 0.025), though still below exact path transport (0.847 ± 0.018).	The present recursive attenuation proxy is now a meaningful constructive baseline: it closes much of the oracle gap, though a stronger strictly local approximation to transported error remains open.
Morphology regime	In the exploratory noise-resilience regime map (Fig. 5), depth-2 trees show their largest shunting gains near $N_I = 0$, whereas depth-3 trees retain their strongest gains around $N_I = 5$.	Morphology appears to shift the inhibitory operating regime in which shunting is most useful, suggesting that conductance normalization and tree geometry interact rather than contributing independently.
Activation choice	In the corrected five-seed frozen activation audit, identity outperforms frozen <code>param_tanh</code> for LocalCA on noise resilience in both architectures (0.590/0.859 vs. 0.109/0.772 for additive at $N_I=0/10$; 0.411/0.769 vs. 0.331/0.701 for shunting), while the backprop effect remains strongly regime-dependent.	The post-voltage firing-rate transform is a real modeling variable, not just an implementation detail. Even a fixed saturating nonlinearity changes the operating point substantially, so the main paper’s mechanistic claims should be read in the learned bounded <code>param_tanh</code> regime used throughout the headline experiments.

Table S2: **Which components carry the observed gains?** A compact summary of the empirical division of labor across architecture, broadcast design, and LocalCA rule variants.

Core	Broadcast	Decoder	Test (mean $\pm$ std)
Shunting	per-soma	local	0.912 $\pm$ 0.005
Shunting	per-soma	backprop	0.909 $\pm$ 0.008
Shunting	local-mismatch	local	0.146 $\pm$ 0.046
Shunting	local-mismatch	backprop	0.146 $\pm$ 0.037
Additive	per-soma	local	0.894 $\pm$ 0.007
Additive	per-soma	backprop	0.900 $\pm$ 0.001
Additive	local-mismatch	local	0.342 $\pm$ 0.058
Additive	local-mismatch	backprop	0.348 $\pm$ 0.095

Table S3: **Local-mismatch recheck (MNIST, 5F)**. Per-soma is consistently strong; local-mismatch remains much weaker.

Dataset	BP ceiling	Best local (5F)	Gap
MNIST	0.971	0.914 $\pm$ 0.003	5.7%
Fashion-MNIST	0.889	0.838 $\pm$ 0.003	5.1%
Context gating	0.861	0.803 $\pm$ 0.006	5.8%

Table S4: **Local competence**. Backprop ceilings are matched-capacity shunting references from dedicated 5-seed standard-training refreshes for MNIST and context gating and from the dedicated 5-seed Fashion-MNIST competence sweep. Local values are 5F with per-soma broadcast on shunting dendritic cores. Context gating additionally uses an HSIC auxiliary objective (weight 0.01; Appendix E). Error bars on local values: ± 1 s.d. across 5 seeds.

A Supplementary Results

Mechanistic Support and Broadcast Controls

Rule-Family Ranking

Which Components Carry?

Broadcast Mode Comparison and Local-Mismatch Recheck

Competence, Verification, and Stress Tests

Phase 1 Capacity Ceilings

Local Competence and Regime Dependence

Ablation Results

Extended Gradient Analysis and N_I Sweep Detail

Verification and Reproducibility

Additional Stress Tests

FA/DFA Baseline Comparison

CIFAR-10 Results

Table S6 provides the exact values for the main-text CIFAR-10 figure. In this stronger compact E/I family, shunting performs well under standard training (49.5% vs. 46.6% for additive), so the harder-data stress test no longer unfairly penalizes the shunting architecture. The broadcast-fidelity ladder also survives once decoder-aware soma mapping is used: additive LocalCA rises from 32.6% under rank-1 per-soma broadcast to 42.3% under transported error, while shunting rises from 28.8% under per-soma to 34.7% with random low-rank $K=4$ and 47.4% with transported error, nearly matching the 49.5% shunting backprop ceiling.

Condition	Metric	Value
Decoder: local vs backprop vs frozen (MNIST, sandbox)	test acc	0.379 vs 0.379 vs 0.176
Shunting vs additive, matched (MNIST, sandbox)	Δ test	+0.210
Per-soma, path on vs off (synthetic task, sandbox)	test / MI(E,I;C)	-0.003 / +0.017

Table S5: **Mechanistic ablations.** Results from controlled small-network sandbox experiments using minimal architectures to isolate individual factors. Sandbox runs use a small 2-layer dendritic network (branch factor [9]) trained for 30 epochs; they are not directly comparable to the headline capacity-calibrated results.

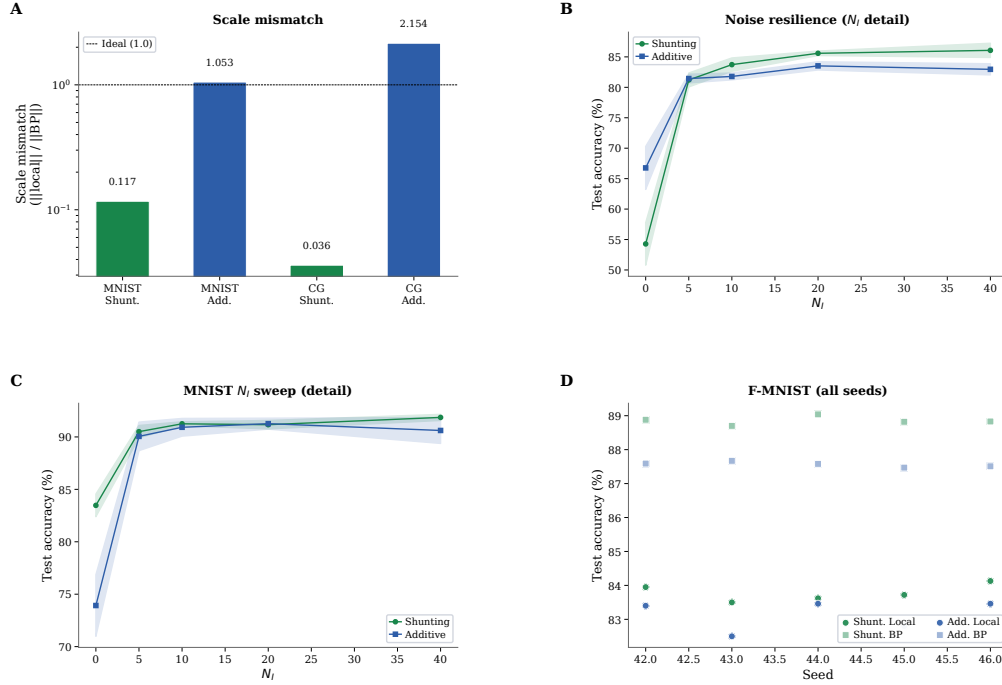


Figure S2: **Extended gradient and N_I analysis (supplementary).** (A) Scale mismatch bars: shunting achieves near-ideal scale (0.117); additive exhibits order-of-magnitude distortion (> 1.0). (B) Noise-resilience N_I dose-response with error bands (± 1 s.d.). (C) MNIST N_I dose-response detail with error bands. (D) Fashion-MNIST individual seeds for all conditions, showing consistency across runs.

Condition	Model	Test accuracy
standard	additive	46.6 ± 1.5
standard	shunting	49.5 ± 0.7
LocalCA per-soma	additive	32.6 ± 0.6
LocalCA per-soma	shunting	28.8 ± 0.8
LocalCA path transport	additive	42.3 ± 1.2
LocalCA path transport	shunting	47.4 ± 0.6
shunting low-rank	$K=4$	34.7 ± 1.1

Table S6: **Exact CIFAR-10 values for the corrected strong-family mechanism extension (5 seeds, validation-best epoch).** This table gives the exact numbers for the main-text harder-data ladder in Fig. 6. The key technical detail is that LocalCA maps output error back to decoder-input/soma space through the decoder Jacobian when the decoder is nonlinear; without that correction, the same family gave misleading near-chance results for per-soma and path transport. With the corrected rule, transported error again acts as a strong upper bound, nearly matching the stronger shunting backprop ceiling, while naive rank-1 per-soma broadcast still underperforms and random low-rank feedback closes part of that gap.

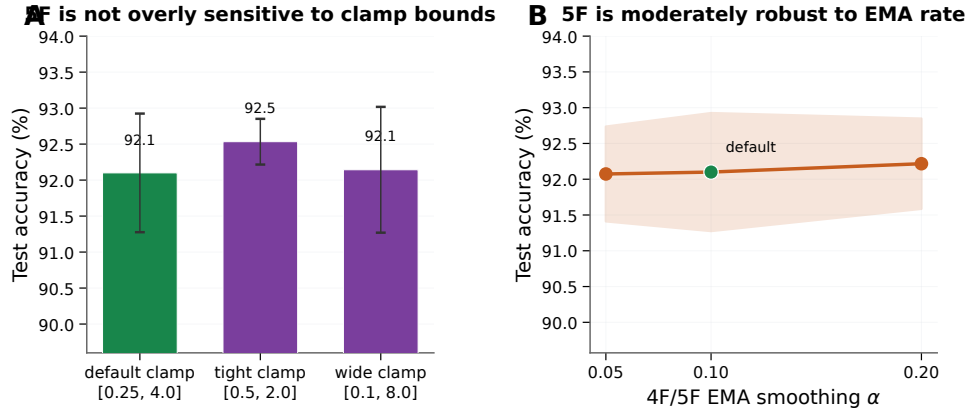


Figure S3: **5F sensitivity analysis (MNIST, shunting, per-soma broadcast)**. (A) Sensitivity to the ϕ_n clamp interval. The default clamp [0.25, 4.0] is already strong ($92.1 \pm 0.8\%$); tightening it to [0.5, 2.0] slightly improves performance ($92.5 \pm 0.3\%$), while widening it to [0.1, 8.0] is effectively neutral ($92.1 \pm 0.9\%$). (B) Sensitivity to the 4F/5F EMA rate used for online statistics. Varying α from 0.05 to 0.20 changes mean test accuracy by less than 0.2 pp. Together these results suggest that the 5F gain is not driven by a brittle single hyperparameter choice, even though the factor itself remains heuristic. Errors: ± 1 s.d. across 3 seeds.

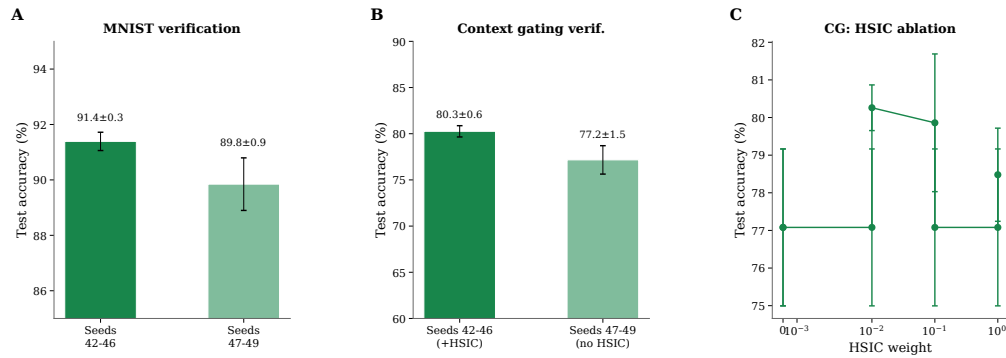


Figure S4: **Verification and reproducibility (supplementary)**. (A) MNIST verification: main seeds (42–46) yield $91.4 \pm 0.3\%$; held-out seeds (47–49) yield $89.8 \pm 0.9\%$, confirming generalization. (B) Context gating verification: main seeds (+HSIC) $80.3 \pm 0.6\%$; held-out seeds (no HSIC) $77.2 \pm 1.5\%$. The roughly 3 pp gap reflects HSIC removal, not seed sensitivity. (C) HSIC weight ablation on context gating: moderate weights (0.01–0.1) perform best.

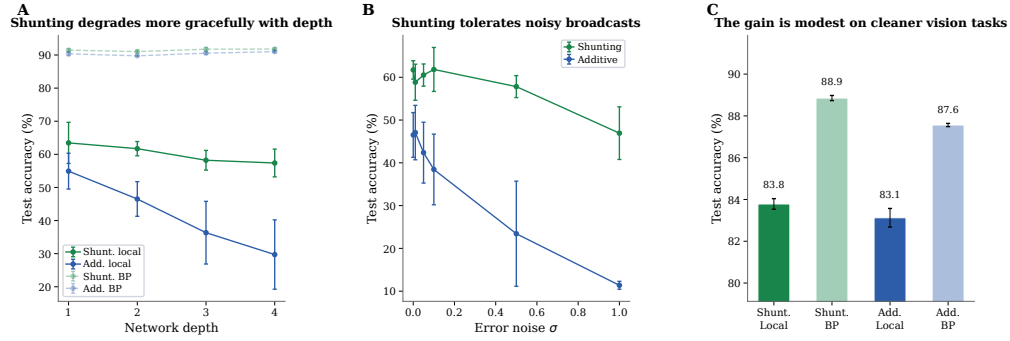


Figure S5: **Additional stress tests.** (A) Depth scaling: shunting local (green) degrades more gracefully from 63.5% to 57.4%; additive local (blue) drops from 54.9% to 29.7%. Backprop ceilings (dashed) remain stable. (B) Broadcast-noise robustness: shunting remains stable for $\sigma \leq 0.1$; additive degrades rapidly and reaches chance at $\sigma = 1$. (C) Fashion-MNIST: the shunting gain is modest on this cleaner benchmark (83.8% local vs. 88.9% backprop), consistent with the regime-dependent story in the main text.

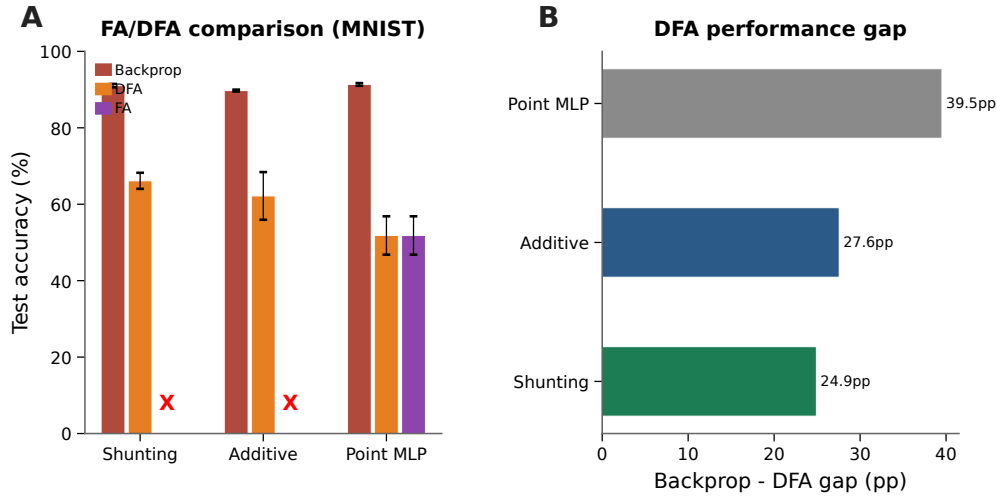


Figure S6: **Feedback alignment baselines (MNIST).** (A) Grouped comparison: standard backprop (gray), DFA (orange), and FA (purple). Red X marks indicate FA failure on dendritic architectures (dimensional incompatibility between random feedback matrices and block-structured dendritic layers). DFA achieves 66.1% on shunting vs. 62.2% additive vs. 51.8% point MLP. (B) Backprop-DFA gap: dendritic architectures (24.9–27.6 pp) show smaller gaps than point MLPs (39.5 pp), suggesting conductance-based architecture is partially compatible with random feedback.

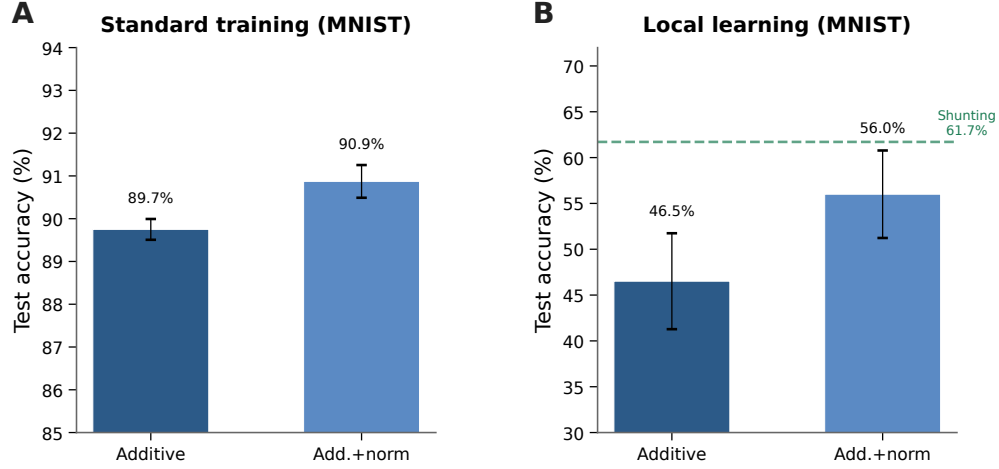


Figure S7: **Additive + normalization control (MNIST)**. (A) Standard backprop: normalization provides a small boost (89.7% $\rightarrow$ 90.9%). (B) Local learning: normalization improves additive from 46.5% to 56.0% (+9.5 pp), partially closing the gap to shunting (61.7%, dashed green). The remaining gap (−5.7 pp) indicates that shunting provides benefits beyond simple voltage normalization.

Additive Normalization Control

Structured Feedback, Activation, and Exploratory Regimes

Morphology $\times$ Inhibition Regime Map

The main-text morphology figure shows the exploratory regime map, while the appendix keeps the summary tables and architecture details that go with it. We continue to treat it as exploratory support for the claim that morphology changes the inhibitory operating regime in which shunting is most useful; a fuller mechanistic account would require matched theory diagnostics for each cell in the grid.

Random Low-Rank Broadcast Channels

Frozen Activation-Choice Audit

B Implementation Details

Biological Plausibility Assumptions

Each synapse has access to: presynaptic activity x_j (local), compartment voltage V_n (local membrane potential), reversal potential E_j (fixed by receptor type), and total input resistance $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$ (computable from local conductances). The 4F modulator ρ_n requires online estimation of voltage covariance (biologically plausible via slow calcium signals); the 5F factor ϕ_n requires a linear regression proxy (implementable via eligibility traces). The only non-local quantity is the broadcast error e_n , which requires a top-down or neuromodulatory signal from the soma to dendritic compartments. We assume per-neuron resolution (one error per output neuron) for the per-soma broadcast mode.

Dendritic Structure in Code

We use the morphology notation $[b_1, b_2, \dots, b_D]$ for rooted dendritic trees, where each factor gives the fan-in from one dendritic stage to the next on the path to the soma. For example, $[3, 3]$ means that each soma aggregates 3 proximal branches and each proximal branch aggregates 3 distal branches, for 9 distal leaves per soma; $[3, 3, 3]$ adds a third dendritic stage with 27 leaves per soma. In code, each excitatory unit is instantiated as a `DendriNet`, which is a stack of `DendriticBranchLayers`. Each branch layer contains an excitatory synapse bank, an optional inhibitory synapse bank, a

Setting	Broadcast	Test accuracy
<i>Noise resilience, shunting, corrected rank bridge</i>		
per-soma	rank-1 per-soma	0.662 ± 0.016
path propagation	recursive attenuation proxy	0.762 ± 0.047
low-rank	$K=1$	0.453 ± 0.126
low-rank	$K=2$	0.630 ± 0.042
low-rank	$K=4$	0.764 ± 0.025
low-rank	$K=8$	0.795 ± 0.019
path transport	exact conductance-stage transport	0.847 ± 0.018
<i>Cue routing, learned shunting router, corrected rank/structure sweep</i>		
per-soma	tuned rank-1	0.803 ± 0.125
low-rank	$K=1$	0.742 ± 0.022
low-rank	$K=2$	0.846 ± 0.082
low-rank	$K=4$	0.748 ± 0.085
pathway-vector	tuned structured rank-2	0.815 ± 0.126

Table S7: **Corrected rank bridge and rank-structure comparison.** On shunting noise resilience, both the current path-propagation proxy and higher-rank random low-rank broadcast close a substantial fraction of the gap between rank-1 per-soma and exact path transport, with low-rank continuing to improve up to $K=8$. On cue routing, the corrected learned-router sweep shows that moving beyond rank-1 helps, but the best current unstructured low-rank setting ($K=2$) is still slightly better than the tuned structured pathway-vector variant. We therefore interpret routed credit assignment as an open higher-rank problem rather than as a solved pathway-vector rescue.

branch-to-parent aggregation map, and an optional post-voltage reactivation. In the main feed-forward models studied here, `somatic_synapses=false`, so synaptic inputs terminate on dendritic branches rather than directly on the soma. Likewise, `input_mode=1` together with nonzero `ie_synapses_per_branch_per_layer` means that inhibition enters each dendritic branch through explicit inhibitory synapses even when we do not instantiate a separate inhibitory neuron population (`inhibitory_layer_sizes=[]`). The additive and shunting cores are therefore architecture-matched at the tree level; they differ in the branch-level voltage integration rule rather than in whether a dendritic tree exists. In all reported dendritic runs, synaptic and dendritic conductance parameters are pushed through positive transforms (`softplus` in the main paper runs), so the conductances themselves remain nonnegative at every optimization step; unconstrained decoder weights are separate readout parameters and are not interpreted as conductances.

Post-Voltage Reactivation

The conductance stage produces a pre-activation compartment voltage V_n , after which each branch layer may apply a monotone scalar nonlinearity. The codebase supports identity (none), fixed `tanh/sigmoid/relu`, and several parametric families including `param_tanh`, `param_tanh_only_m`, and piecewise linear-saturating variants. The main manuscript sweeps use `param_tanh`, a learnable bounded firing-rate transform of the form $(\tanh(m(V - b)) + 1)/2$, together with an explicitly configured nonnegative transfer stage (ReLU in the corrected positive-conductance runs) so that the first synaptic drive remains nonnegative even after input corruption or structured task noise. Most headline feedforward sweeps use a linear decoder, while the CIFAR extension uses a nonlinear decoder and therefore maps output error back to decoder-input/soma space through the decoder Jacobian before applying LocalCA. Unless explicitly overridden, additive and shunting cores can start from different reactivation initializations in code; activation-fairness ablations therefore match additive initialization to the shunting setting when comparing alternative reactivation choices directly. The appendix activation audit additionally fixes `param_tanh` after initialization to isolate activation shape from activation plasticity.

Dataset / strategy	Core	N_I	Identity (none)	Frozen param_tanh
MNIST / LocalCA	additive	0	0.780 ± 0.014	0.709 ± 0.037
MNIST / LocalCA	additive	10	0.907 ± 0.003	0.881 ± 0.005
MNIST / LocalCA	shunting	0	0.841 ± 0.007	0.778 ± 0.020
MNIST / LocalCA	shunting	10	0.905 ± 0.009	0.840 ± 0.016
Noise resilience / LocalCA	additive	0	0.590 ± 0.162	0.109 ± 0.006
Noise resilience / LocalCA	additive	10	0.859 ± 0.001	0.772 ± 0.009
Noise resilience / LocalCA	shunting	0	0.411 ± 0.028	0.331 ± 0.020
Noise resilience / LocalCA	shunting	10	0.769 ± 0.004	0.701 ± 0.006
MNIST / backprop	additive	0	0.925 ± 0.001	0.942 ± 0.001
MNIST / backprop	additive	10	0.921 ± 0.001	0.950 ± 0.001
MNIST / backprop	shunting	0	0.963 ± 0.0004	0.950 ± 0.001
MNIST / backprop	shunting	10	0.967 ± 0.001	0.946 ± 0.001
Noise resilience / backprop	additive	0	0.896 ± 0.003	0.106 ± 0.007
Noise resilience / backprop	additive	10	0.893 ± 0.002	0.927 ± 0.002
Noise resilience / backprop	shunting	0	0.898 ± 0.001	0.869 ± 0.002
Noise resilience / backprop	shunting	10	0.926 ± 0.001	0.912 ± 0.001

Table S8: **Frozen activation-shape audit on matched dendritic architectures (5 seeds per condition)**. Identity (none) and a frozen bounded post-voltage reactivation param_tanh lead to markedly different operating regimes even after correcting the task and enforcing nonnegative first-layer synaptic drive. In this corrected audit, identity is the safer LocalCA choice on both MNIST and the noise-resilience task, while the backprop effect is strongly regime-dependent: frozen param_tanh helps additive most clearly in the inhibited noise-resilience setting ($N_I=10$) but collapses it at $N_I=0$. These runs match additive reactivation initialization to the shunting setting and hold the activation parameters fixed, isolating activation shape from both the earlier additive/shunting initialization asymmetry in code and from activation learning itself.

Dataset	Core	N_I	Identity (none)	Frozen param_tanh
MNIST	additive	0	CV 0.280, cosine 0.220	CV 1.122, cosine 0.074
MNIST	additive	10	CV 0.310, cosine 0.212	CV 1.079, cosine 0.094
MNIST	shunting	0	CV 0.393, cosine 0.170	CV 0.770, cosine 0.114
MNIST	shunting	10	CV 0.385, cosine 0.281	CV 1.177, cosine 0.106
Noise resilience	additive	0	CV 0.124, cosine 0.079	CV 0.114, cosine 0.052
Noise resilience	additive	10	CV 0.414, cosine 0.206	CV 0.840, cosine 0.094
Noise resilience	shunting	0	CV 0.255, cosine 0.324	CV 0.344, cosine 0.341
Noise resilience	shunting	10	CV 0.762, cosine 0.140	CV 2.497, cosine 0.079

Table S9: **Frozen activation-shape theory audit on matched LocalCA runs (5 seeds per condition)**. Each entry reports conductance-stage path-gain dispersion (CV) and direct per-soma compartment-error fidelity (cosine between the per-soma broadcast and the exact pre-reactivation compartment error). The corrected positive-input theory audit largely agrees with the empirical one: freezing param_tanh usually broadens path-gain dispersion and weakens per-soma fidelity, especially in the inhibited settings where the main mechanistic story lives. The one mild exception is low-inhibition shunting on noise resilience, where frozen param_tanh leaves fidelity roughly unchanged while still hurting final accuracy, suggesting that activation choice affects optimization and representation as well as direct credit fidelity.

536 Units and Parameterization

537 Decoder Update Modes

538 W_{dec} maps $V_L \rightarrow \hat{y}$. Modes: **backprop** ($\nabla_W L$ via autograd), **local** ($\Delta W = \eta \langle \delta_0 V_L^T \rangle_B$), **frozen**
539 ($\Delta W = 0$).

Quantity	Symbol	Convention
Voltage	V	Normalized to $[-1, 1]$
Conductances	$g^{\text{syn}}, g^{\text{den}}$	Nonneg. via softplus
Leak conductance	g^{leak}	Set to 1
Input resistance	R^{tot}	≤ 1

Table S10: Units and normalization.

Category	Implemented modes	Role in this paper
Core models	point MLP; dendritic MLP; dendritic additive; dendritic shunting;	Main comparisons use dendritic additive vs. dendritic shunting; routed cores are used for cue integration.
Training strategies	pathway-routed dendritic cores standard backprop; LocalCA; DFA; FA; transported-broadcast oracle	Main text uses standard and LocalCA; DFA/FA remain appendix baselines; path transport is an oracle upper bound.
Broadcast modes	scalar; per-soma; local mismatch; low-rank; path transport; pathway vector	Main text centers on per-soma, low-rank, and transported-error controls; scalar and local-mismatch are supporting controls, while pathway vector remains an appendix structured-feedback extension.
Recent dendritic extensions	path propagation; router-derived branch roles; pathway-vector feedback; inhibitory homeostasis	These were added to test morphology-aware credit transport and structured branch-specific plasticity beyond the original 3F/4F/5F family.

Table S11: Implemented model families and LocalCA operating modes. Only a subset is used for the main claims; the remaining modes serve as controls, ablations, or structured extensions.

540 Model and Mode Inventory

541 Representative Manuscript Architectures

542 Effective Broadcast Dimensionality in Headline Architectures

543 LocalCA Extension Compatibility

544 Seed Protocol and Claim Status

545 The seed depth is not uniform across every sweep, and we therefore separate headline evidence from
546 exploratory screening. The main mechanistic gradient-fidelity sweep uses 5 seeds per condition, the
547 corrected path-transport oracle sweep uses 5 seeds per condition, the cue-routing seed-repeat and
548 tuning sweeps use 5 seeds per condition, the dedicated Fashion-MNIST competence sweep uses 5
549 seeds per condition, the corrected flattened-CIFAR-10 mechanism extension in Table S6 also uses 5
550 seeds per condition, and the frozen activation audit in Table S8 uses 5 seeds per condition. A smaller
551 set of support sweeps, including the 5F sensitivity and morphology $\times$ inhibition map, use 3 seeds per
552 condition and are described as exploratory where appropriate. By contrast, early architecture-selection
553 or stress-test screens with only 1–2 seeds are treated as exploratory and are either omitted from the
554 main claims or explicitly framed as pilots. For transparency, the BP ceilings in Table S4 for MNIST
555 and context gating now come from dedicated matched-capacity 5-seed standard-training refreshes
556 rather than the earlier 2-seed calibration pass.

557 Compute Resources

558 All experiments were run on an internal GPU cluster. Most MNIST, Fashion-MNIST, and synthetic-
559 task runs used a single NVIDIA A100 or H100 GPU and finished in minutes to tens of minutes per
560 seed; the flattened CIFAR-10 runs required longer single-GPU jobs but still did not use distributed or
561 multi-GPU training. Reported headline metrics are aggregated over 5 seeds unless noted otherwise;
562 this now includes the corrected flattened-CIFAR-10 mechanism extension. The remaining 3-seed

Setting	Encoder	E layers	Tree	Synapses / branch	Seeds	Notes
Gradient fidelity / path-gain / noise-resilience mechanism	id.+ReLU	[128, 128]	[3, 3]	$N_E=40, N_I \in \{0, 5, 10, 20, 40\}$	5 fidelity, 5 oracle	Main mechanistic sweeps in Figs. 2 and 3; dendritic reactivation is <code>param_tanh</code> .
MNIST / context-gating local competence	id.+ReLU	[128]	[3, 3]	$N_E=40, N_I=20$	5	Best local 5F per-soma runs used in Table S4; dendritic reactivation is <code>param_tanh</code> .
Fashion-MNIST competence	id.+ReLU	[128]	[3, 3]	$N_E=40, N_I=20$	5	Dedicated five-seed standard-vs-local competence sweep; dendritic reactivation is <code>param_tanh</code> .
Cue-routing PV-LocalCA	router+ReLU [64]		[2]	$N_E=10, N_I=4$	5	Learned router with two pathway groups and rank-2 structured feedback; dendritic reactivation is <code>param_tanh</code> .
CIFAR-10 compact E/I mechanism extension	id.+ReLU	[20] E, [20] I	[3, 3, 3, 3]	$N_E=25, N_{EI}=25, N_{IE}=25$	5	Harder-data extension with explicit inhibitory population, decoder [32, 16], and decoder-aware soma-error mapping for LocalCA.
Morphology×inhibition appendix map	id.+ReLU	[128, 128]	[2, 2], [3, 3], [4, 4], [2, 2, 2], [3, 3, 3]	$N_E=40, N_I \in \{0, 5, 10, 20, 40\}$	3	Exploratory support sweep; dendritic reactivation is <code>param_tanh</code> .

Table S12: **Representative architectures used in the manuscript.** All dendritic rows use explicit branch-level excitatory and inhibitory synapse banks with `somatic_synapses=false`, nonnegative conductances, and a nonnegative first-layer drive (implemented by an explicit ReLU transfer stage in the corrected positive-conductance runs). The tree column gives the dendritic morphology per excitatory neuron, while the synapse counts report excitatory and inhibitory synapses per branch.

563 results are confined to selected exploratory support sweeps, and the broader exploratory screens used
564 to tune and diagnose the method required additional compute beyond the final reported runs.

565 Code, Data, and Asset Availability

566 MNIST, Fashion-MNIST, CIFAR-10, and MICrONS are public datasets or public benchmark assets
567 cited in the paper; we use them under their standard published access conditions and cite their original
568 sources in the bibliography. The present anonymous submission does not include a public code URL
569 in order to preserve double-blind review, but the paper specifies the model family, training modes, key
570 hyperparameters, and representative configurations, and we plan to release code and configuration
571 files after review. The paper does not release a new dataset or a stand-alone pretrained model asset.

Setting	Hidden width(s)	Output dim	Effective default per-soma field
Gradient fidelity / path-gain / noise resilience	128, 128	10	Scalar / rank-1 at both dendritic layers
MNIST / Fashion-MNIST competence	128	10	Scalar / rank-1 at the dendritic hidden layer
Context gating competence	128	2	Scalar / rank-1 at the dendritic hidden layer
Cue routing	64	2	Scalar / rank-1 by default; higher-rank tests use explicit low-rank or pathway-vector feedback
Compact CIFAR-10 harder-data extension	20	decoder-input dim 20	Vector-valued per-soma after decoder-aware mapping; higher-rank tests use explicit low-rank or path transport

Table S13: **Effective dimensionality of the default “per-soma” broadcast in the headline architectures.** In code, per-soma uses the output error vector only when layer width matches decoder output dimension; otherwise it falls back to a shared scalar. The main feedforward settings in this paper therefore test a genuinely rank-1 broadcast regime by default, which is why the higher-rank low-rank, path-transport, and pathway-vector controls are important.

Extension	Meaning	Compatible with	Typical use here
Per-soma broadcast	Somatic error vector when widths match, otherwise scalar fallback	3F, 4F, 5F	Default main-text LocalCA setting; in the headline feedforward runs this is effectively rank-1
Low-rank broadcast	Small number of broadcast channels instead of a scalar or full field	3F, 4F, 5F	Intermediate control when rank-1 is too restrictive
Path transport	Recursive tree-transported error using conductance-stage path gains	3F, 4F, 5F	Oracle upper-bound broadcast
Path propagation	Approximate path-gain attenuation applied to broadcast	3F, 4F, 5F	Morphology-aware modifier for deeper trees
Pathway vector	Structured branch/pathway-specific broadcast field	3F, 4F, 5F	Used mainly in PV-LocalCA on routed tasks
Branch-role rules	Scales updates by router- or pathway-derived branch specialization	3F, 4F, 5F	Used in the structured pathway extension
Inhibitory homeostasis	Auxiliary local inhibitory balancing term	3F, 4F, 5F	Optional stabilizer in routed or shunting settings

Table S14: Compatibility of the newer LocalCA extensions with the base 3F/4F/5F family. These extensions mostly modify the error field or branch scaling rather than defining a new “6F” or “7F” rule.

572 Algorithm

Algorithm 1 Local Credit Assignment

- 1: **Input:** Model, batch (x, y) , config $\mathcal{C}$
- 2: Forward pass; loss L , output error δ^y
- 3: Somatic/core error $\delta_0 = \text{decoder-input Jacobian product } J_{\text{dec}}(h_{\text{core}})^\top \delta^y$ (or $W_{\text{dec}}^\top \delta^y$ for a linear decoder)
- 4: **for** each layer n (reverse) **do**
- 5: $e_n = \text{broadcast}(\delta_0, \mathcal{C})$; optionally gate by pathway roles
- 6: Compute ρ_n, ϕ_n (EMA estimators, if enabled)
- 7: Apply the configured local update (Eq. 6 or 7)
- 8: **end for**
- 9: Clip gradients; optimizer step

Rule	Factors	Cost	Best regime
3F	$x, (E - V), e$	$\mathcal{O}(1)$	Baseline
4F	$3F + \rho$	$\mathcal{O}(1)$	Improved conditioning
5F	$4F + \phi$	$\mathcal{O}(d_n)$	Strongest overall
5F+VH	5F + local voltage homeostasis	$\mathcal{O}(d_n)$	Stabilized LocalCA / warm-up
PV-LocalCA	3F + structured e_n + role alignment	$\mathcal{O}(r_n)$	Pathway discovery

Table S15: Variant taxonomy. ‘5F+VH’ denotes the same base 5F rule with an additional strictly local voltage-centering auxiliary that can also be scheduled through multi-stage LocalCA warm-up; the codebase also supports this as a shorthand ‘5f_vh’ alias in configuration files.

Component	Analog	Interpretation
R_n^{tot}	Input resistance	Sensitivity modulation
$(E_j - V_n)$	Synaptic driving force	Local gradient factor
Shunting	Divisive normalization	$\partial V / \partial g_I \propto -V$
ρ_n	Layer relevance	Output correlation
ϕ_n	Signal propagation	Conditional predictability
Pathway vector	Branch-specific top-down drive	Structured teaching signal
Role alignment	Pathway-selective plasticity	Branch specialization

Table S16: Biological analogs.

C Theoretical Details

Variant Taxonomy

Random-Broadcast Alignment

Proposition 3 (Supportive random-broadcast alignment). *Let the broadcast matrix B_n have i.i.d. zero-mean entries with $\mathbb{E}[B_n^\top B_n] = \alpha I$. Then the expected cosine between the resulting local gradient and the exact gradient is positive, $\mathbb{E}[\cos \angle(g^{\text{local}}, g^{\text{exact}})] \geq c_n > 0$, by an argument analogous to feedback alignment [9]. The constant c_n depends on how strongly local eligibility factors correlate with the conductance-stage path gain in Eq. (4); shunting improves this correlation by narrowing the spread of intermediate sensitivities.*

We treat this argument as supportive rather than central. The main theoretical object in the paper is still the exact pre-activation compartment error $\partial L / \partial V_n$; under identity upward transfer this reduces to $\alpha_n \delta_0$, while with post-voltage activation the corresponding effective path gain additionally includes local $f'_n(V_n)$ factors. The main empirical question is how faithfully different broadcast fields approximate that quantity.

Biological Analogs

D Morphology-Aware Extensions

Path-integrated propagation. Modulate broadcast error by $\pi_n = \pi_{n-1} \cdot R_{n-1}^{\text{tot}} \cdot \bar{g}_{n-1}^{\text{den}}$, approximating depth attenuation from Eq. 4.

Depth modulation. Per-branch scaling $\rho_j = \rho_{\text{base}} / (d_j + \alpha)$, mirroring cable attenuation.

Dendritic normalization. $\Delta g_j^{\text{den}} \leftarrow \Delta g_j^{\text{den}} / (\sum_k g_k^{\text{den}} + \varepsilon)$, analogous to homeostatic scaling [21].

Pathway-vector broadcast. For tasks with latent pathway structure, the broadcast becomes a low-dimensional vector over router-inferred pathway roles. This vector is gated by local pathway activity and propagated through the block-structured tree so that earlier branches receive feedback aligned with their downstream descendants.

Condition	MNIST		Fashion-MNIST	
	Exc σ	Inh σ	Exc σ	Inh σ
BP + Shunting	0.94±0.01	0.87±0.01	0.92±0.01	0.84±0.01
BP + Additive	1.30±0.01	1.22±0.01	1.16±0.01	1.05±0.01
Local + Shunting	1.22±0.01	1.71±0.02	1.13±0.06	1.36±0.06
Local + Additive	1.69±0.16	1.57±0.07	1.43±0.16	1.11±0.12
Song et al. 2005	$\sigma = 0.94$ (rat V1 L5, EPSP amplitude)			
MICrONS E→E	$\sigma = 1.14$ (mouse V1, synapse size, $n=3.9M$)			
MICrONS I→E	$\sigma = 0.92$ (mouse V1, synapse size, $n=3.5M$)			

Table S17: **Weight distribution width.** Log-normal σ of excitatory and inhibitory synaptic weights (mean±std across 5 seeds). Biological references are included only as descriptive anchors. With biologically constrained E/I ratios (4:1), shunting self-normalization produces weight heterogeneity in the range spanned by those references, whereas additive integration is broader and less stable.

Router-derived branch roles. Each branch receives a role profile inferred from router assignments and incoming block weights rather than from a hand-coded apical/basal label. Plasticity is then modulated by branch selectivity and a synapse-specific role-alignment factor, emphasizing pathway-consistent updates without imposing a heuristic branch taxonomy.

E HSIC Auxiliary Objectives

Following [17], we use kernel-based HSIC objectives. Self-decorrelation: $\mathcal{L}^{\text{self}} = B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Z \mathbf{H})$. Target-correlation: $\mathcal{L}^{\text{target}} = -B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Y \mathbf{H})$. Moderate weights (0.01–0.1) help on context gating; negligible on MNIST. Online statistics (ρ_n, ϕ_n) use Welford’s algorithm [20].

F Online Variant with Eligibility Traces

Continuous-time eligibility: $\tau_e \dot{e}_j^{\text{syn}} = -e_j^{\text{syn}} + x_j(E_j - V_n)R_n^{\text{tot}}$. Update: $\Delta g_j^{\text{syn}} \propto \int e_j^{\text{syn}}(t)m_n(t) dt$ [18, 19].

G Weight Statistics and Biological References

As a secondary analysis, we examine whether the conductance-based update naturally induces characteristic weight statistics. Because synaptic updates scale with input resistance $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$, the voltage equation implies multiplicative weight dynamics, creating a regime analogous to exponentiated-gradient updates. Such multiplicative dynamics suggest log-normal weight distributions in trained networks [38, 39]. We compare the resulting distributions against two biological references: electrophysiological data from Song et al. [38] (rat V1 L5 pyramidal, $\sigma = 0.94$) and the MICrONS mm³ connectome [40] (mouse V1 synapse sizes).

Model weight distributions are log-normal. We fit log-normal distributions to the active (top- K masked) excitatory synaptic weights after training on MNIST and Fashion-MNIST. Table S17 shows the fitted σ (standard deviation of $\log w$) for each condition. Shunting consistently produces narrower distributions than additive integration, and backpropagation produces narrower distributions than local learning. These two factors combine approximately additively: BP+Shunting → narrowest, Local+Additive → broadest.

Shunting is closest to the biological references considered here. BP+Shunting produces $\sigma = 0.94$ on MNIST, very close to the Song et al. reference ($\Delta = 0.008$). Local+Shunting produces $\sigma = 1.13$ – 1.22 , close to MICrONS E→E synapse sizes ($\sigma = 1.14$; $\Delta \leq 0.08$). By contrast, Local+Additive produces $\sigma = 1.43$ – 1.69 , broader than any biological reference considered. This pattern is consistent across datasets (MNIST and Fashion-MNIST), suggesting it reflects structural properties of the learning rule rather than task-specific features.

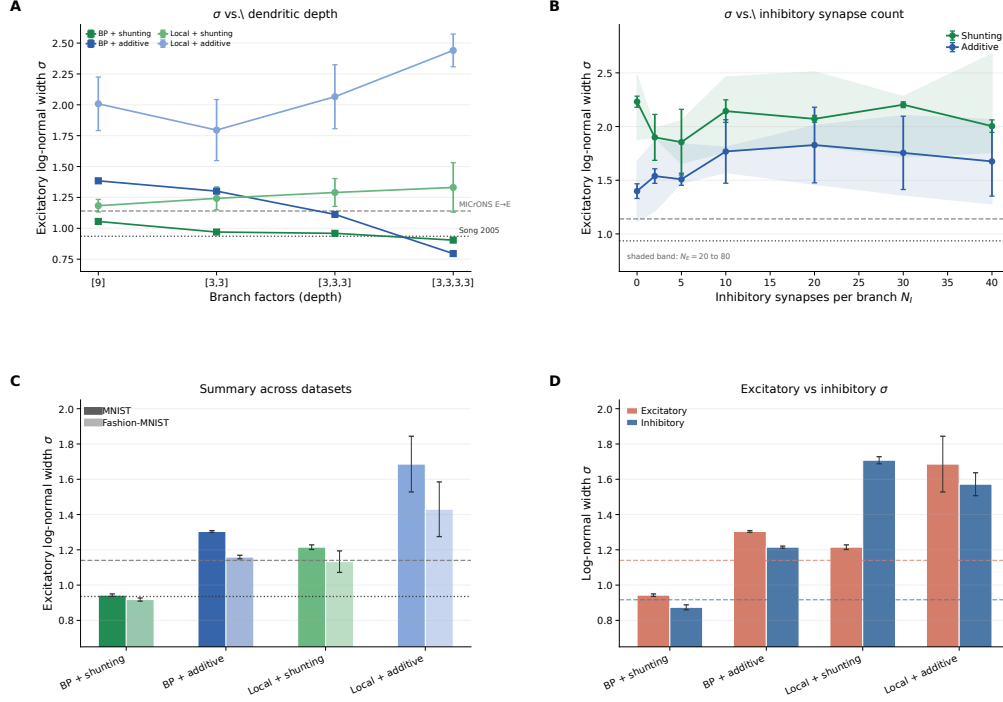


Figure S8: **Synaptic weight distribution sensitivity to architecture.** (A) Log-normal σ vs. dendritic depth (branch factors [9] to [3,3,3,3]). Both backprop and local shunting remain closer to the biological-width references than their additive counterparts across depth. (B) σ vs. inhibitory synapses per branch (N_I). Without inhibition ($N_I=0$), all conditions are overly broad; with biologically constrained $N_I=10$ –20, shunting enters the biological range. Shaded bands show the sweep over $N_E=20$ to 80. (C) Summary of MNIST and Fashion-MNIST results with biological references. The shunting ordering is consistent across datasets. (D) Excitatory vs. inhibitory weight σ on MNIST with MICrONS E $\rightarrow$ E and I $\rightarrow$ E reference bands.

The match depends on inhibitory conductance. We test sensitivity to the E/I synapse ratio using a sweep over $N_I \in \{0, 2, 5, 10, 20, 30, 40\}$ inhibitory synapses per branch (Appendix Fig. S8). Without inhibition ($N_I=0$), all conditions produce $\sigma > 2.0$, far from biology. With biologically constrained ratios ($N_I=10$ –20, corresponding to E/I ratios of 4:1 to 2:1; [42]), shunting networks enter the biological range ($\sigma \approx 0.9$ –1.2), while the effect of N_I on additive networks is less systematic. The weight distribution shape is also robust to dendritic depth (branch factors [9] through [3,3,3,3]; see Appendix Fig. S8A): BP+Shunting converges toward Song et al. with increasing depth, while Local+Shunting remains near MICrONS E $\rightarrow$ E across depths.

H Weight Distribution Sensitivity Analysis

MICrONS connectome analysis. We extracted synapse size distributions from the MICrONS mm³ condensed connectome (v1181; 9.0M connections, 71.9K neurons; 89% excitatory). Splitting by pre/post cell type: E $\rightarrow$ E connections ($n=3.85$ M) show $\sigma_{\text{size}}=1.14$ and CV=1.01; I $\rightarrow$ E ($n=3.53$ M) show $\sigma=0.92$ and CV=1.03; E $\rightarrow$ I ($n=1.11$ M) show $\sigma=0.83$; I $\rightarrow$ I ($n=0.51$ M) show $\sigma=0.85$. Inhibitory connections are more frequently multi-synaptic (I $\rightarrow$ E: 37.5% multi-synapse; E $\rightarrow$ E: 11.5%), and inhibitory synapses are on average smaller (median 3532 vs. 4676 for E $\rightarrow$ E).

Sensitivity to ee_synapses. Using the E/I grid sweep ($N_E \in \{20, 40, 80\} \times N_I \in \{0, 2, 5, 10, 20, 30, 40\}$; 168 models), excitatory σ increases modestly with N_E in both architectures (+0.4–0.6 from $N_E=20$ to 80), but the shunting-additive ordering is preserved at all N_E values. The dominant driver of σ is N_I : going from $N_I=0$ to $N_I=10$ reduces shunting σ by approximately 1.0–1.5 units, while the effect on additive is smaller and less systematic.

Sensitivity to depth. From the depth-scaling sweep ([9], [3,3], [3,3,3], [3,3,3,3]; 80 models, 5 seeds each), BP+Shunting excitatory σ decreases with depth ($1.06 \rightarrow 0.90$), converging toward Song et al. (0.94). Local+Shunting σ increases slightly ($1.18 \rightarrow 1.33$) but remains in the biological range at shallow depths. BP+Additive shows the strongest depth sensitivity ($1.38 \rightarrow 0.79$), while Local+Additive diverges with depth ($2.01 \rightarrow 2.44$). The morphology-aware modulators (path propagation, depth/centrality weighting) have negligible effect on σ .

Mechanistic interpretation. The shunting conductance $g_n^{\text{tot}} = g_n^{\text{leak}} + \sum g^{\text{exc}} + \sum g^{\text{inh}}$ appears in the update factor $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$. As inhibitory conductance increases, g_n^{tot} grows and R_n^{tot} shrinks, compressing the multiplicative scale of weight updates. This is analogous to a learning rate that self-adjusts via the total conductance: strong synapses contribute more to g_n^{tot} , reducing their own future update magnitude. This acts as a natural form of weight-dependent regularization. The resulting equilibrium distribution is log-normal, consistent with Loewenstein et al. [41] and related multiplicative spine-dynamics analyses. Additive integration lacks this self-normalizing mechanism, producing broader distributions that fall outside the biological range.

I Synthetic Benchmark Descriptions

Each synthetic task is generated procedurally from a fixed random seed and uses fixed train/val/test splits for reproducibility. The MNIST-derived synthetic tasks use flattened 28×28 inputs; cue integration instead uses a small vector input with two output classes. Output dimensionality therefore varies by task and is stated in the task description when it differs from 10-way digit classification.

Context gating. Inputs are contextual figure-ground MNIST examples. The right half of the image carries the original MNIST digit, while the left half is replaced by low-amplitude uniform noise; all pixels therefore remain nonnegative. This task tests whether credit assignment can exploit a simple context-dependent spatial structure without leaving the nonnegative-drive regime of the conductance model. The HSIC auxiliary objective (weight 0.01) is used to decorrelate layer representations, adding roughly 3 pp under local learning.

Noise resilience. Inputs are flattened MNIST images corrupted by structured channel noise: a fixed 50-dimensional latent Gaussian perturbation is projected into input space and added to each example, with $\sigma_{\text{task}} = 1.5$, then clipped to $[0, 1]$. The projection is fixed across training, so the network must filter correlated interference to recover the underlying class signal. Credit assignment must remain coherent under the corrupted signal, making this task particularly sensitive to path-gain dispersion.

Info shunting. Inputs contain two additive streams: a low-frequency signal and a high-frequency distractor drawn with equal energy, together with a separate context channel indicating which stream is relevant on each trial. In the current code this benchmark uses signed synthetic inputs, so it violates the nonnegative presynaptic-drive assumption enforced in the corrected headline experiments. We therefore retain it only as a legacy exploratory task and exclude it from the corrected main-text claims.

Cue integration. Two noisy cue streams (A and B) are presented simultaneously, each carrying partial information about one of 2 classes. A binary one-hot context signal indicates which cue is reliable on each trial (cue A or cue B), and each cue vector is clipped to $[0, 1]$ after noise is added. The fixed-pathway variant duplicates the context into both cue branches; the learned-routing variant requires the router to discover the cue separation from data. This task is used in Sec. 4.9 to test structured pathway-vector broadcast while staying inside the nonnegative-input regime.

J Depth Scaling and Noise Robustness

Depth scaling. Varying dendritic depth from 1–4 layers (branch factors [9] to [3, 3, 3, 3]): shunting local degrades from 63.5% to 57.4%; additive local degrades more steeply from 54.9% to 29.7%. The shunting advantage grows with depth ($+8.5 \rightarrow +27.7$ pp). Backprop ceilings remain stable at about 90–92%. See Fig. S5A.

696 **Noise robustness.** Gaussian noise $\mathcal{N}(0, \sigma^2)$ on broadcast error: shunting is robust to $\sigma \leq 0.1$ (about
697 62%) while additive drops from 46.5% to chance at $\sigma=1.0$, confirming that shunting credit signals
698 carry useful learning information beyond the broadcast magnitude. See Fig. S5B.

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